

Amazon Route 53

**AWS Certified Solutions Architect Associate
Certification SAA-C03**

What is DNS?

DNS stands for **Domain Name System**. It is often called the "phonebook of the internet."

What does DNS do?

DNS translates **human-readable domain names** (like `www.google.com`) into **IP addresses** (like `142.250.182.4`) that computers use to identify each other on the network.

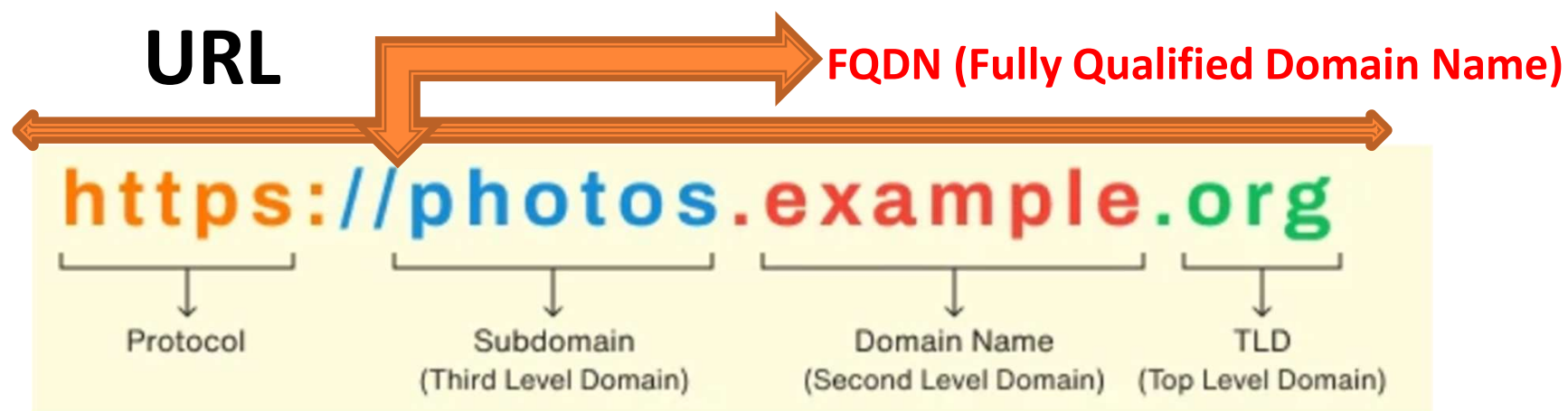
Why is DNS important?

Humans remember domain names, not IP addresses. Without DNS, you'd have to type IP addresses to visit websites.

DNS uses hierarchical naming structure :- `.com` `example.com`
`www.example.com` `api.example.com`

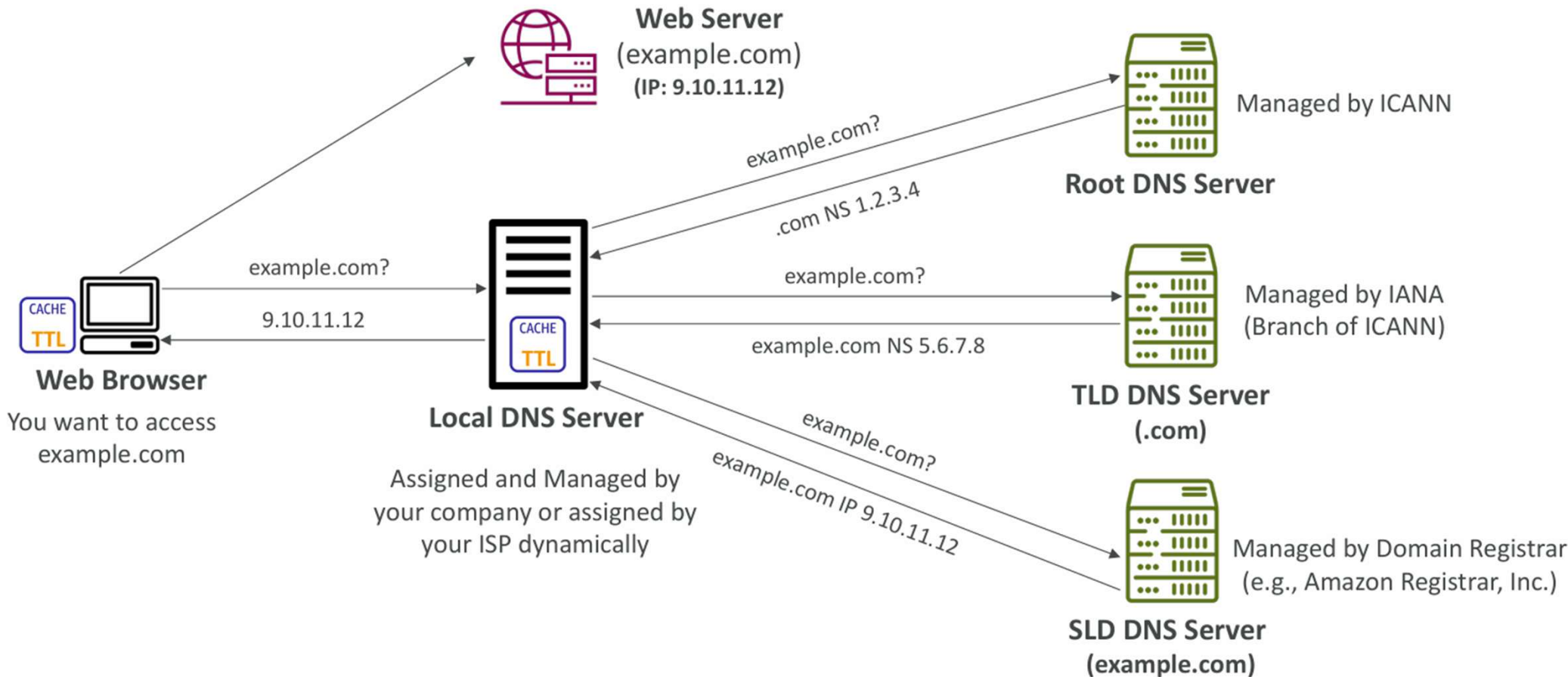
DNS Terminologies

- ▶ Domain Registrar: Amazon Route 53, GoDaddy, ...
- ▶ DNS Records: A, AAAA, CNAME, NS, ...
- ▶ Zone File: contains DNS records
- ▶ Name Server: resolves DNS queries (Authoritative or Non-Authoritative)
- ▶ Top Level Domain (TLD): .com, .us, .in, .gov, .org, ...
- ▶ Second Level Domain (SLD): amazon.com, google.com, ...



<https://www.youtube.com/@techwithtwinkle>

How DNS Works



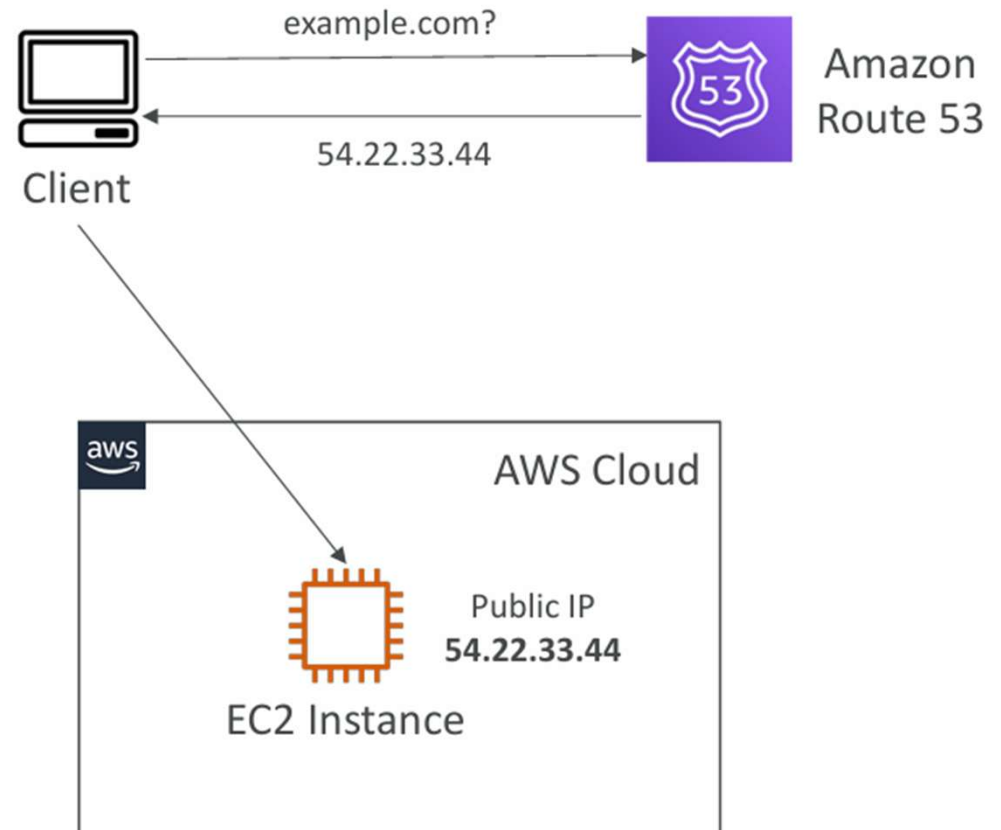
Amazon Route 53

- ▶ **Amazon Route 53** is a **highly available and scalable cloud Domain Name System (DNS) web service**.
- ▶ **Key Functions of Route 53:**
- ▶ **Domain Registration** – You can buy and manage domain names (e.g., example.com) directly through Route 53.
- ▶ **DNS Routing** – Routes user requests to internet applications hosted in:
 - AWS (like EC2, S3, CloudFront)
 - On-premises infrastructure
- ▶ **Health Checks & Failover** – It monitors the health of resources and routes traffic away from unhealthy endpoints.
- ▶ **Routing Policies** – Supports:
 - Simple routing
 - Weighted routing
 - Latency-based routing
 - Geo DNS
 - Failover routing
 - Multi-value answer routing

- ▶ A highly available, scalable, fully managed and Authoritative DNS
- ▶ Authoritative = the customer (you) can update the DNS records
- ▶ Route 53 is also a Domain Registrar
- ▶ Ability to check the health of your resources
- ▶ The only AWS service which provides 100% availability SLA

Why Route 53?

53 is a reference to the traditional DNS port



Route 53 – Records

How you want to route traffic for a domain

- ▶ **Each record contains:**
- ▶ **Domain/subdomain Name** – e.g., example.com
- ▶ **Record Type** – e.g., A or AAAA
- ▶ **Value** – e.g., 12.34.56.78
- ▶ **Routing Policy** – how Route 53 responds to queries
- ▶ **TTL** – amount of time the record cached at DNS Resolvers
- ▶ Route 53 supports the following DNS record types:
- ▶ (must know) A / AAAA / CNAME / NS
- ▶ (advanced) CAA / DS / MX / NAPTR / PTR / SOA / TXT / SPF / SRV

Route 53 – Record Types

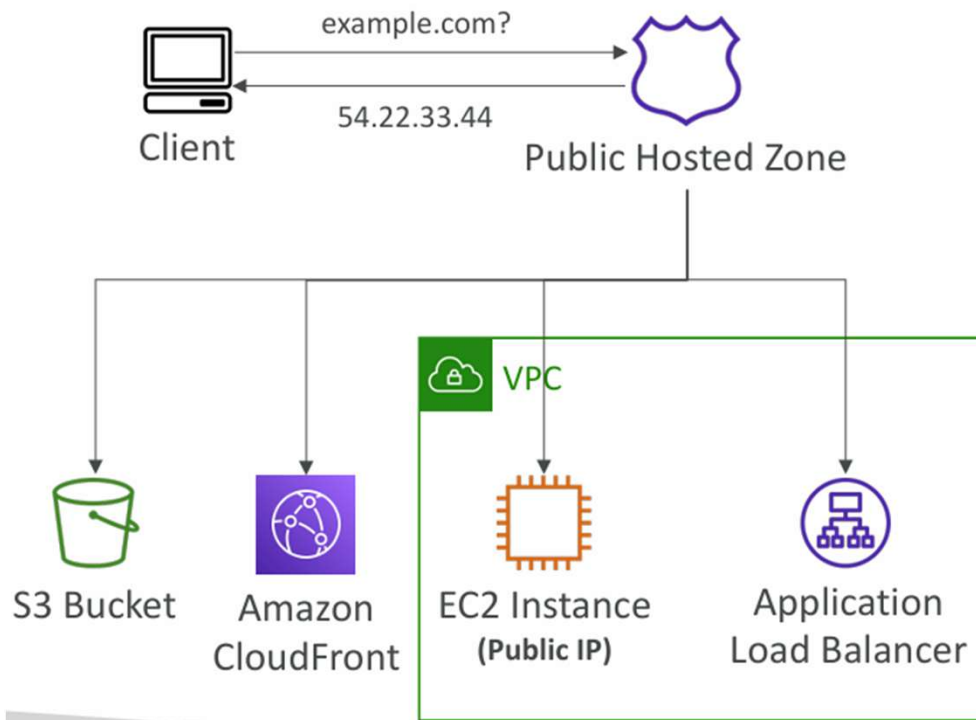
- ▶ A – maps a hostname to IPv4
- ▶ AAAA – maps a hostname to IPv6
- ▶ CNAME – maps a hostname to another hostname
- ▶ The target is a domain name which must have an A or AAAA record
- ▶ Can't create a CNAME record for the top node of a DNS namespace (Zone Apex)
- ▶ Example: you can't create for example.com, but you can create for www.example.com
- ▶ NS – Name Servers for the Hosted Zone
- ▶ Control how traffic is routed for a domain

Route 53 – Hosted Zones

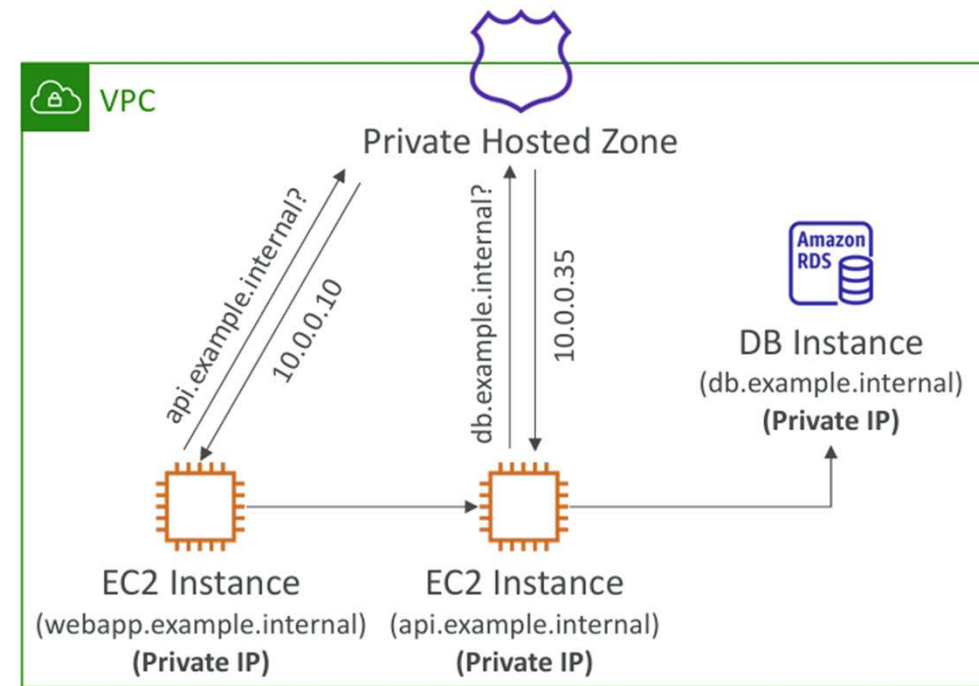
- ▶ A container for records that define how to route traffic to a domain and its subdomains
- ▶ Public Hosted Zones – contains records that specify how to route traffic on the Internet (public domain names) application1.mypublicdomain.com
- ▶ Private Hosted Zones – contain records that specify how you route traffic within one or more VPCs (private domain names) application1.company.internal
- ▶ You pay \$0.50 per month per hosted zone

Route 53 – Public vs. Private Hosted Zones

Public Hosted Zone

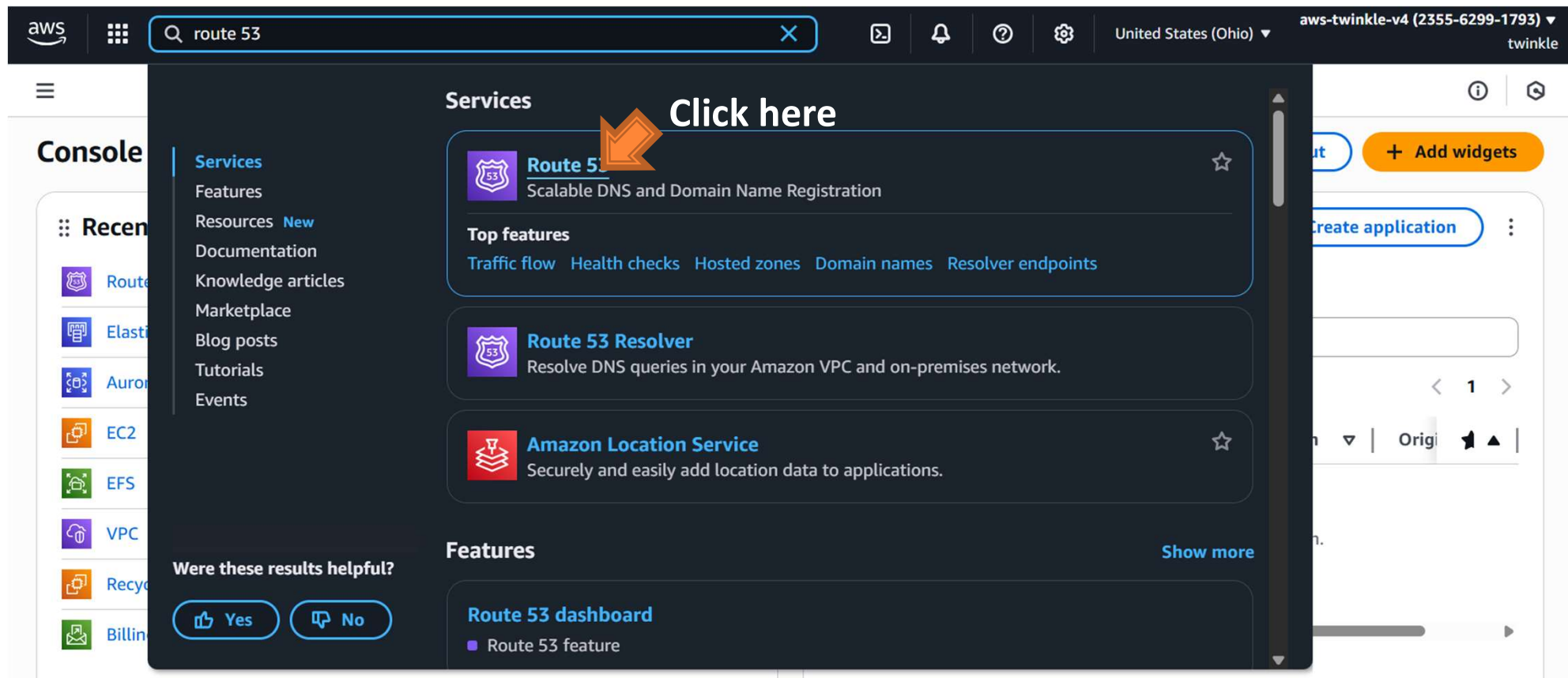


Private Hosted Zone



Route 53 – Registering a Domain

- ▶ Go to search console → search Route 53



<https://www.youtube.com/@techwithtwinkle>

Amazon Route 53

A reliable way to route users to internet applications

Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service.

Get started with Route 53

Get started by registering a domain, configuring DNS, or using another Route 53 feature.

Get started

Pricing (US)

[View pricing](#)

Click on get started

How it works

<https://www.youtube.com/@techwithtwinkle>

More resources

The screenshot shows the Amazon Route 53 console with the 'Get started' tab selected. The 'Register a domain' option is highlighted with a blue border and a radio button. An orange arrow points to this option. The other options are 'Transfer domain', 'Create hosted zones', 'Configure health checks', 'Configure traffic flow', and 'Configure resolvers'. The page also shows the AWS logo, a search bar, and the user's account information (aws-twinkle-v4 (2355-6299-1793)).

Click on get started

Get started

Register domains [Info](#)

Pricing for domain names varies by top-level domain (TLD). For more information, view [price with different TLDs](#).

Search for domain

Check availability for a domain

techtwinklesharma.com



Search

Search domain



Selected domains (0/5)

Search for domains and make a selection

Proceed to checkout

Standard pricing

Pricing for domain names varies by top-level domain (TLD), such as .com or .org.

Search result

Domain	Price/year	Actions
techtwinklesharma.com Exact match	15.00 USD Renews at 15.00 USD	Select

Select this It will charge 15 USD for domain
If the price will change it will show u here



Check availability for a domain

techtwinklesharma.com



Search

Standard pricing

Pricing for domain names varies by top-level domain (TLD), such as .com or .org.

Search result

Domain	Price/year	Actions
techtwinklesharma.com Exact match	15.00 USD Renews at 15.00 USD	Selected

Suggested available domains (10)

You can register up to five domains at a time.

Domain	Price/year	Actions
--------	------------	---------

Selected domains (1/5)

Domain registration fee

techtwinklesharma.com

15.00 USD

Renews at 15.00 USD

Remove

Subtotal:

15.00 USD

The domain registration fee displayed is for 1 year. You can change duration on the next page.

Proceed to checkout



Click on proceed to checkout

<https://www.youtube.com/@techwithtwinkle>

- Step 1
● **Pricing**
- Step 2
○ Contact information
- Step 3
○ Review and submit

Pricing [Info](#)

Domain pricing options

Domain name

techtwinklesharma.com

Duration (price)

1 year (15.00 USD)

Auto-renew

☒ On



Auto-renew is turned on for 1 domain.

We will send an email to the registrant contact before expiration to remind you that auto-renew is currently turned on. You can turn it off at any time by using the Route 53 console. For more information, see [Turning off Auto-renew for a Domain](#).

Subtotal: 15.00 USD

Applicable taxes will be calculated at checkout.

Click on Next

Next

Here is your domain name

Auto renew you can select & deselect its your choice

Select the duration

- Step 1
● Pricing
- Step 2
● **Contact information**
- Step 3
○ Review and submit

Contact information [Info](#)

Registrant contact

General information

Contact type

Organization

First name

Last name

Email

Phone number

<https://www.youtube.com/@techwithtwinkle>

Fill the contact info

Tech contact

☒ Same as the registrant contact

Billing contact

☒ Same as the registrant contact

Privacy protection

☒ Turn on privacy protection for all contacts of techtwinklesharma.com
When you turn on privacy protection, contact information is hidden from WHOIS queries.

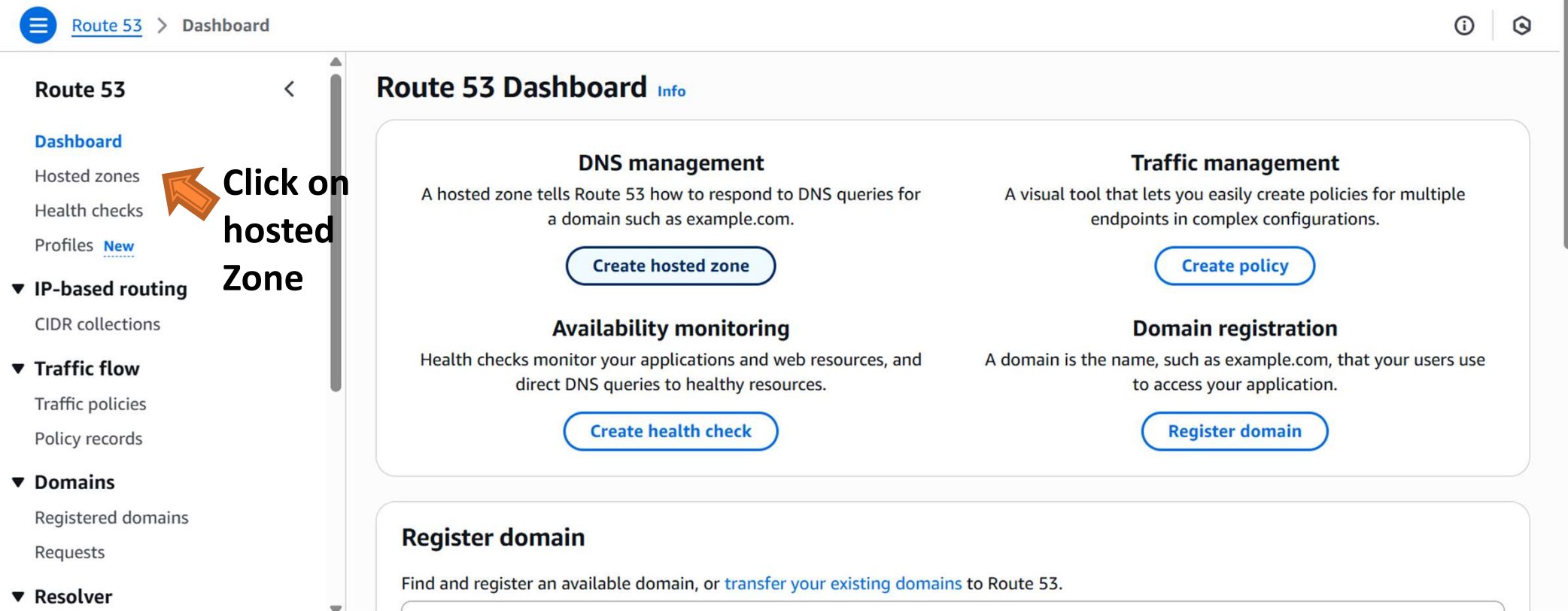
Click on Next



**After that it will show you review page, do payment aftrre that click
on submit
After few time domain will be available**

<https://www.youtube.com/@techwithtwinkle>

Create First Record in Route 53



The screenshot shows the AWS Route 53 Dashboard. On the left is a navigation sidebar with the following items: 'Route 53' (selected), 'Dashboard', 'Hosted zones', 'Health checks', 'Profiles' (with a 'New' button), 'IP-based routing' (expanded), 'Traffic flow', 'Domains' (expanded), and 'Resolver'. An orange arrow points from the text 'Click on hosted Zone' to the 'Hosted zones' link in the sidebar. The main content area is titled 'Route 53 Dashboard' and contains four cards: 'DNS management' (with a 'Create hosted zone' button), 'Traffic management' (with a 'Create policy' button), 'Availability monitoring' (with a 'Create health check' button), and 'Domain registration' (with a 'Register domain' button). Below these is a 'Register domain' section with a description and a link to transfer existing domains.

Route 53 > Dashboard

Route 53 Dashboard Info

DNS management
A hosted zone tells Route 53 how to respond to DNS queries for a domain such as example.com.
[Create hosted zone](#)

Traffic management
A visual tool that lets you easily create policies for multiple endpoints in complex configurations.
[Create policy](#)

Availability monitoring
Health checks monitor your applications and web resources, and direct DNS queries to healthy resources.
[Create health check](#)

Domain registration
A domain is the name, such as example.com, that your users use to access your application.
[Register domain](#)

Register domain
Find and register an available domain, or [transfer your existing domains](#) to Route 53.

Click on hosted Zone

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ IP-based routing

- CIDR collections

▼ Traffic flow

- Traffic policies
- Policy records

▼ Domains

- Registered domains
- Requests

▼ Resolver

Public **techtwinklesharma.online** Info

Delete zone Test record Configure query logging

► Hosted zone details Edit hosted zone

Records (3) DNSSEC signing Hosted zone tags (0)

Records (3) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias

1

☐	Record ... ▼	Type ▼	Routin... ▼	Differ... ▼	Alias ▼	Value/Route traffic to ▼	TTL (s...
☐	techtwink...	NS	Simple	-	No	ns-1521.awsdns-62.org. ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk	172800

Click on create record



Route 53 > Hosted zones > techtwinklesharma.online > Create record

Create record Info

Quick create record

▼ Record 1

Record name Info

test .techtwinklesharma.online

Keep blank to create a record for the root domain.

Alias

Record type Info

A – Routes traffic to an IPv4 address and some AWS resources

Value Info

11.22.33.44

Enter multiple values on separate lines.

Switch to wizard

Delete

Enter subdomain as you wish

Select record type A

Enter any value of your choice

Value Info

11.22.33.44

Enter multiple values on separate lines.

TTL (seconds) Info

300

Recommended values: 60 to 172800 (two days)

Routing policy Info

Simple routing

Add another record

Click on create record

Create records

View existing records

The following table lists the existing records in techtwinklesharma.online.

<https://www.youtube.com/@techwithtwinkle>

Route 53

Dashboard

Hosted zones

Health checks

Profiles [New](#)

IP-based routing

CIDR collections

Traffic flow

Traffic policies

Policy records

Domains

Registered domains

Requests

Resolver

Record for techtwinklesharma.online was successfully created.

Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.

View status

Records (4) Info



Delete record

Import zone file

Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type

Routing p...

Alias

1



☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1867.awsdns-41.co.uk
☐	test.techtwinklesharma.online	A	Simple	-	No	ns-1521.awsdns-62.org.
☐	www.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
☐						172.31.43.104

Record is created now

Route 53 – EC2 Setup

aws

Search [Alt+S]

Asia Pacific (Mumbai) ▼ aws-twinkle-v4 (2355-6299-1793) ▼ twinkle

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

...

Compute

Amazon Elastic Compute Cloud (EC2)

Create, manage, and monitor virtual servers in the cloud.

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.

Click on Launch instance

Launch a virtual server

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

[View dashboard](#)

<https://www.youtube.com/@techwithtwinkle>

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough

Do not show me this message again.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

MvInstance

Add additional tags

Enter name of instance

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0144277607031eca2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

<https://www.youtube.com/@techwithtwinkle>

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0144277607031eca2 (64-bit (x86), uefi-preferred) / ami-01b04c8bf1b7e7e07 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250808.1 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-0144277607031eca2

Publish Date

2025-08-08

Username

ec2-user

Info

Verified provider

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0144277607031eca2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't

Cancel

Launch instance

Preview code

Select linux

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value ▼

[Create new key pair](#)

Network settings [Info](#)

Network [Info](#)

vpc-0a506c34e326aff4f

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

Proceed without key pair

Create new security group

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0144277607031eca2

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Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't

[Cancel](#)

[Launch instance](#)

[Preview code](#)

<https://www.youtube.com/@techwithtwinkle>

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Check this

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0144277607031eca2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't

[Cancel](#)

[Launch instance](#)

[Preview code](#)

EC2 > Instances > Launch an instance

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage Info Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Advanced details Info

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-0144277607031eca2

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't)

Cancel Launch instance Preview code

Click on Advanced details

EC2 > Instances > Launch an instance

Allow tags in metadata Info
Select

User data - optional Info
Upload a file with your user data or enter it in the field.
Choose file

<https://www.youtube.com/@techwithtwinkle>

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
# updated script to make it work with Amazon Linux 2023
CHECK_IMDSV1_ENABLED=$(curl -s -o /dev/null -w "%{http_code}"
http://169.254.169.254/latest/meta-data/)
if [[ "$CHECK_IMDSV1_ENABLED" -eq 200 ]]
then
    EC2_AVAIL_ZONE=$(curl -s http://169.254.169.254/latest/meta-
data/placement/availability-zone)"
else
    EC2_AVAIL_ZONE=$(TOKEN=`curl -s -X PUT "http://169.254.169.254/latest/api/token" -H
"X-aws-ec2-metadata-token-ttl-seconds: 21600" ` && curl -s -H "X-aws-ec2-metadata-token:
```

User data has already been base64 encoded

Copy & paste userdata here

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-0144277607031eca2

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't)

Cancel Launch instance Preview code

After that click on launch instance

Launch another 2 instance with different locations other field same as above

Three
Instances
with
different
Locations

The image displays three screenshots of the AWS Management Console, each showing a list of EC2 instances in a different region. The regions are Asia Pacific (Mumbai), Asia Pacific (Hyderabad), and Asia Pacific (Osaka). Each screenshot shows a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4 DNS. The instances are named MyInstance, Myinstance2, and Myinstance3 respectively. The status of the instances is Running, Running, and Initializing. The instance types are t2.micro, t3.micro, and t2.micro. The availability zones are ap-south-1a, ap-south-1a, and ap-south-1a. The public IPv4 DNS addresses are ec2-13-127-17-56.ap, ec2-13-127-17-56.ap, and ec2-13-127-17-56.ap.

<https://www.youtube.com/@techwithtwinkle>

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
MyInstance	i-0c43c8fdcaf1bf54b	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-13-127-17-56.ap

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Myinstance2	i-0525e4f75b0d71a14	Running	t3.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-13-127-17-56.ap

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Myinstance3	i-0458afba642bb4534	Initializing	t2.micro	1/2 checks failed	View alarms +	ap-south-1a	ec2-13-127-17-56.ap

aws twinkle v4 (2355-6299-1793) Asia Pacific (Mumbai)

EC2 > Instances

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
MyInstance	i-0c43c8fdcaf1bf54b	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-13-127-17-56.ap

Click on Launch load balancer

i-0c43c8fdcaf1bf54b (MyInstance)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0c43c8fdcaf1bf54b	Public IPv4 address 13.127.17.56 open address	Private IPv4 addresses 172.31.47.141
IPv6 address -	Instance state Running	Public DNS ec2-13-127-17-56.ap-south-1.compute.amazonaws.com

aws twinkle v4 (2355-6299-1793) Asia Pacific (Mumbai)

EC2 > Load balancers

Load balancers

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

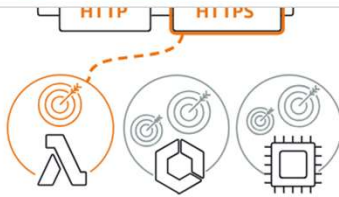
Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
No load balancers						

You don't have any load balancers in ap-south-1

Click on create load balancer

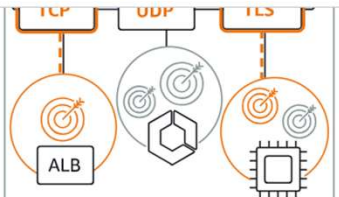
Create load balancer

<https://www.youtube.com/@techwithtwinkle>



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create Application Load Balancer



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

Click on create

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

DemoRoute53ALB

Enter Load balancer name

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Select internet facing

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Select IPv4

<https://www.youtube.com/@techwithtwinkle>

IP pools - new | Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-south-1a (aps1-az1)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-0287ee3f45c829710
IPv4 subnet CIDR: 172.31.32.0/20

☒ **ap-south-1b (aps1-az3)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-0fa41e55155c9f1a7
IPv4 subnet CIDR: 172.31.0.0/20

☒ **ap-south-1c (aps1-az2)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-06b73b9c44595f643
IPv4 subnet CIDR: 172.31.16.0/20

Select All Availability Zones

Security groups | Info
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

☒	launch-wizard-1 sg-0f1cdc929432a734a VPC: vpc-0a506c34e326aff4f
☐	default sg-0f28c11256989da4e VPC: vpc-0a506c34e326aff4f

Listeners and routing | Info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Protocol: HTTP Port: 80

Default action: Forward to Select a target group

1-65535

Remove

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Select launch wizard 1 & deselect default security group

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 [Remove](#)

Protocol

HTTP

Port

80

1-65535

Default action [Info](#)

Forward to [Select a target group](#)

[Create target group](#)

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)

You can add up to 50 more tags.

[Add listener](#)

You can add up to 49 more listeners.

Click on create target group

<https://www.youtube.com/@techwithtwinkle>

► Load balancer tags - optional

Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them. The 'Key' is required, but 'Value' is optional. For example, you can have Key = production-

EC2 > Target groups > Create target group

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Choose instances

Enter target group name

Target group name

demo-tg-route-53

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

Advanced health check settings

Attributes

Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Click on Next

Next

Step 1

Specify group details

Step 2

Register targets

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/1)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone
☒	i-0c43c8fdcaf1bf54b	MyInstance	Running	launch-wizard-1	ap-south-1a

1 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

After that click here

Include as pending below

<https://www.youtube.com/@techwithtwinkle>

Review targets

0 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Filter targets

Show only pending

Remove all pending

< 1 > ⚙

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Label
i-0c43c8fdcaf1bf54b	MyInstance	80	Running	launch-wizard-1	ap-south-1a	172.31.47.141	subnet-0287ee3f45c829710	Aug

1 pending

Click on create target group



aws

Search [Alt+S]

Asia Pacific (Mumbai) aws-twinkle-v4 (2355-6299-1793) twinkl

EC2 > Load balancers > Create Application Load Balancer

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Protocol

HTTP

Port

80

1-65535

Default action

Info

Forward to

Select a target group

Create target group

Refresh Target groups

Remove

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

You can add up to 49 more listeners.

Click on refresh

<https://www.youtube.com/@techwithtwinkle>

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Protocol

Port
1-65535

Default action [Info](#)

Forward to HTTP [Create target group](#)



Select target group

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)

You can add up to 50 more tags.

[Remove](#)

- ap-south-1a [subnet-0287ee3f45c829710](#)
- ap-south-1b [subnet-0fa41e55155c9f1a7](#)
- ap-south-1c [subnet-06b73b9c44595f643](#)

Service integrations [Edit](#)

Amazon CloudFront + AWS Web Application Firewall (WAF): -
AWS WAF: -
AWS Global Accelerator: -

Tags [Edit](#)

-

Attributes

[i](#) Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.

Creation workflow and status

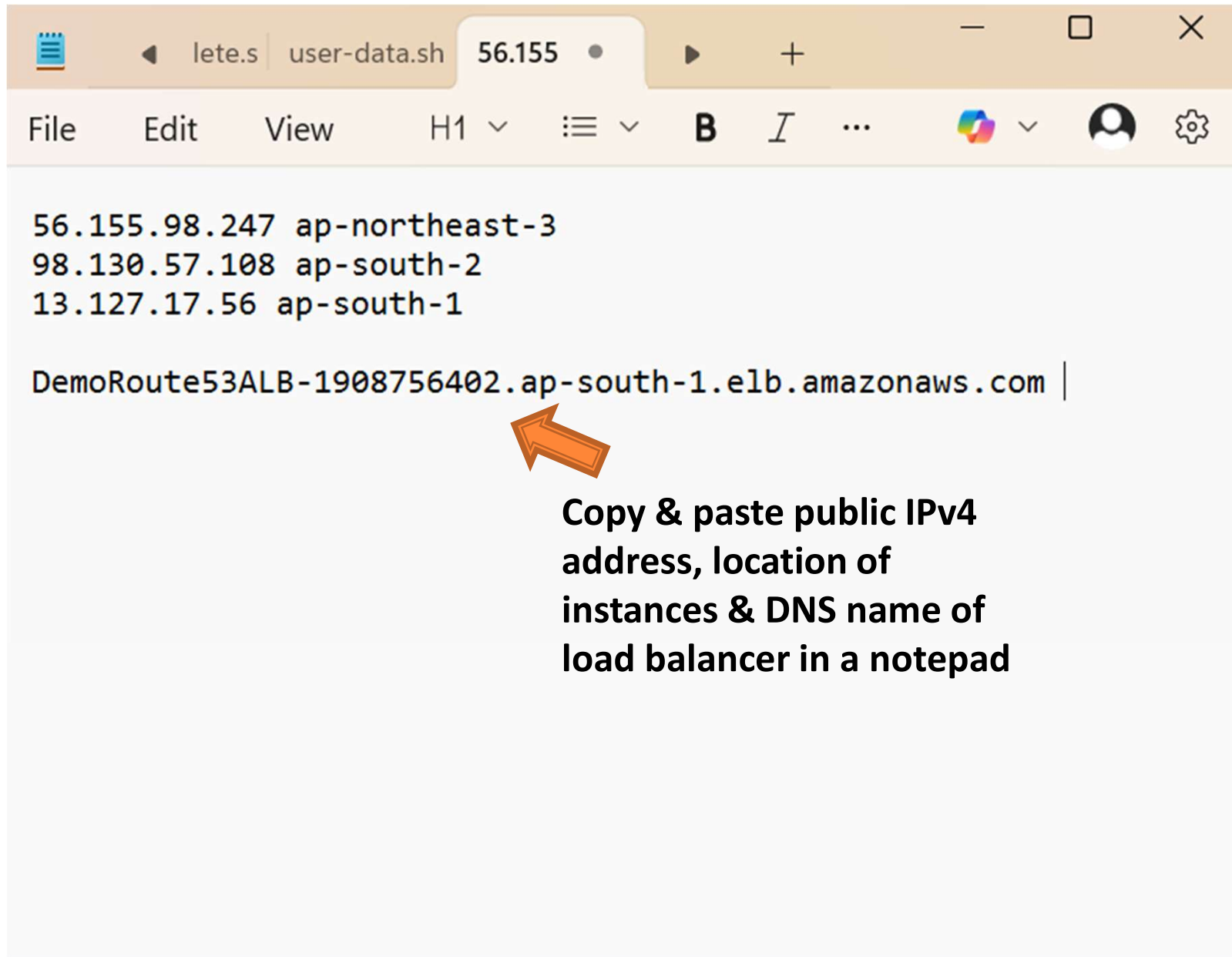
► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

Click on create load balancer



Create load balancer



56.155.98.247 ap-northeast-3
98.130.57.108 ap-south-2
13.127.17.56 ap-south-1

DemoRoute53ALB-1908756402.ap-south-1.elb.amazonaws.com |

Copy & paste public IPv4 address, location of instances & DNS name of load balancer in a notepad

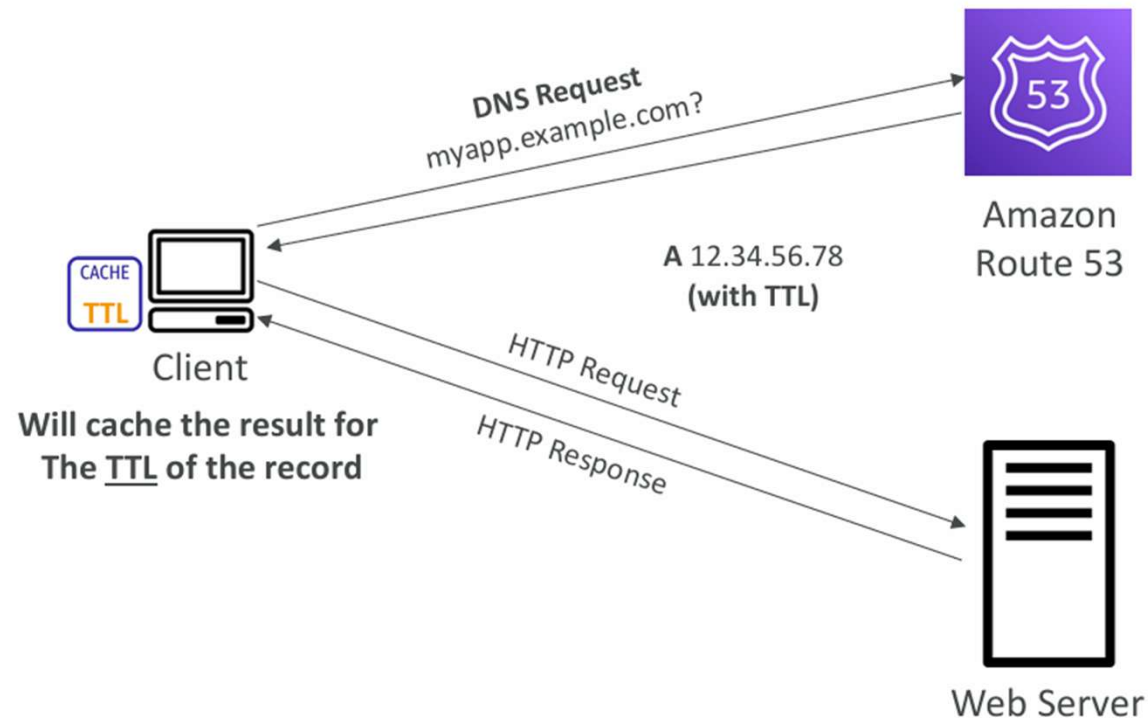
Route 53 – Records TTL (Time To Live)

High TTL – e.g., 24 hr

- Less traffic on Route 53
- Possibly outdated records

Low TTL – e.g., 60 sec.

- More traffic on Route 53 (\$\$)
 - Records are outdated for less time
 - Easy to change records
- ▶ **Except for Alias records, TTL is mandatory for each DNS record**



Route 53 – Records TTL Hands On

The screenshot shows the AWS Route 53 console interface. At the top, a blue notification banner states: "Record for techtwinklesharma.online was successfully created. Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use 'View status' button to check propagation status." Below this, the "Records (4)" section is active, displaying a table of DNS records. An orange arrow points from the text "Click on create record" to the "Create record" button. The left sidebar shows the navigation menu with options like Dashboard, Hosted zones, Health checks, Profiles, IP-based routing, Traffic flow, Domains, and Resolver.

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- ▼ IP-based routing
 - CIDR collections
- ▼ Traffic flow
 - Traffic policies
 - Policy records
- ▼ Domains
 - Registered domains
 - Requests
- ▼ Resolver

Records (4) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type ▼ Routing p... ▼ Alias ▼

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1867.awsdns-41.co.uk
☐	test.techtwinklesharma.online	A	Simple	-	No	ns-1521.awsdns-62.org. a
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
☐	www.techtwinklesharma.online	A	Simple	-	No	172.31.43.104

[Create record](#)

<https://www.youtube.com/@techwithtwinkle>

Create record [Info](#)

Quick create record

[Switch to wizard](#)

▼ Record 1

Delete

Record name [Info](#)

demo .techtwinklesharma.online

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

192.0.2.235

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

Record type [Info](#)

A – Routes traffic to an IPv4 address and some AWS resources

Enter record name

Enter IPv4 address of any instance I am taking ap-south-1 instance IP for now

<https://www.youtube.com/@techwithtwinkle>

Routing policy [Info](#)

demo .techtwinklesharma.online

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

13.127.17.56

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

120

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Simple routing

+1m

1h

1d

Click here 2 times to make TTL for 2 mins

[Add another record](#)

After that click on create records

Create records

Route 53 > Hosted zones > techtwinklesharma.online

Record for techtwinklesharma.online was successfully created.
Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.

Records (5) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	NS	Simple	-	No	ns-1521.awsdns-62.org. ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk.
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-62.org. aws
☐	demo.techtwinklesharma.online	A	Simple	-	No	13.127.17.56
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44

demo.techtwinklesharma.online

Not secure demo.techtwinklesharma.online

Record is created copy & paste this in chrome to check is it working or not

Hello world from ip-172-31-47-141.ap-south-1.compute.internal in AZ ap-south-1a



It is working

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ **IP-based routing**

- CIDR collections

▼ **Traffic flow**

- Traffic policies
- Policy records

▼ **Domains**

- Registered domains
- Requests

▼ **Resolver**

Records (1/5) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by prefix Type Routing policy Alias

	Record name	Type	Routing policy
☒	demo.techtwinklesharma.online	A	Simple
☐	test.techtwinklesharma.online	A	Simple
☐	www.techtwinklesharma.online	A	Simple

Record details

[Edit record](#)

Record name: demo.techtwinklesharma.online

Record type: A

Value: 13.127.17.56

Alias: No

TTL (seconds): 120

Routing policy

Select record

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ **IP-based routing**

- CIDR collections

▼ **Traffic flow**

- Traffic policies
- Policy records

▼ **Domains**

- Registered domains
- Requests

▼ **Resolver**

Records (1/5) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by prefix Type Routing policy Alias

	Record name	Type	Routing policy
☒	demo.techtwinklesharma.online	A	Simple
☐	test.techtwinklesharma.online	A	Simple
☐	www.techtwinklesharma.online	A	Simple

Record details

A – Routes traffic to an IPv4 address and s...

☐ Alias

Value [Info](#)

13.127.17.56

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

120 [1m](#) [1h](#) [1d](#)

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Simple routing

[Cancel](#) [Save](#)

Change this IP

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles **New**

▼ **IP-based routing**

- CIDR collections

▼ **Traffic flow**

- Traffic policies
- Policy records

▼ **Domains**

- Registered domains
- Requests

▼ **Resolver**

Records (1/5) [Info](#)

[Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by priority Type Routing policy Alias

	Record name	Type	Routing policy
☐	techtwinklesharma.online	SOA	Simple
☒	demo.techtwinklesharma.online	A	Simple
☐	test.techtwinklesharma.online	A	Simple
☐	www.techtwinklesharma.online	A	Simple

Value [Info](#)

56.155.98.247

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

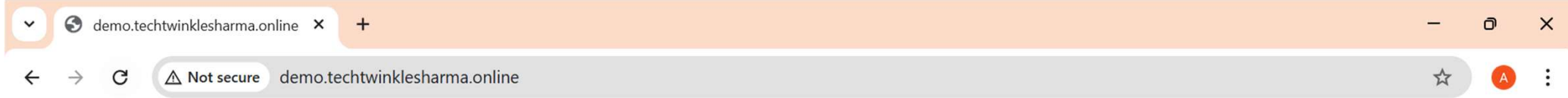
120 [1m](#) [1h](#) [1d](#)

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Simple routing

Click on save  [Save](#) [Save changes](#)

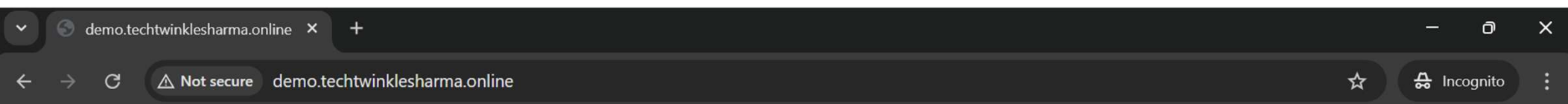


Hello world from ip-172-31-47-141.ap-south-1.compute.internal in AZ ap-south-1a



After changes go to browser & refresh previous page it is showing as previous it will take 2 min to update as TTL

<https://www.youtube.com/@techwithtwinkle>



Hello world from ip-172-31-36-102.ap-northeast-3.compute.internal in AZ ap-northeast-3a



After 2 mins IP Changed

<https://www.youtube.com/@techwithtwinkle>

CNAME vs Alias

AWS Resources (Load Balancer, CloudFront...) expose an AWS hostname:

- ▶ lb1-1234.us-east-2.elb.amazonaws.com and you want myapp.mydomain.com

CNAME:

- Points a hostname to any other hostname. (app.mydomain.com => blabla.anything.com)
- ONLY FOR NON ROOT DOMAIN (aka. something.mydomain.com)

Alias:

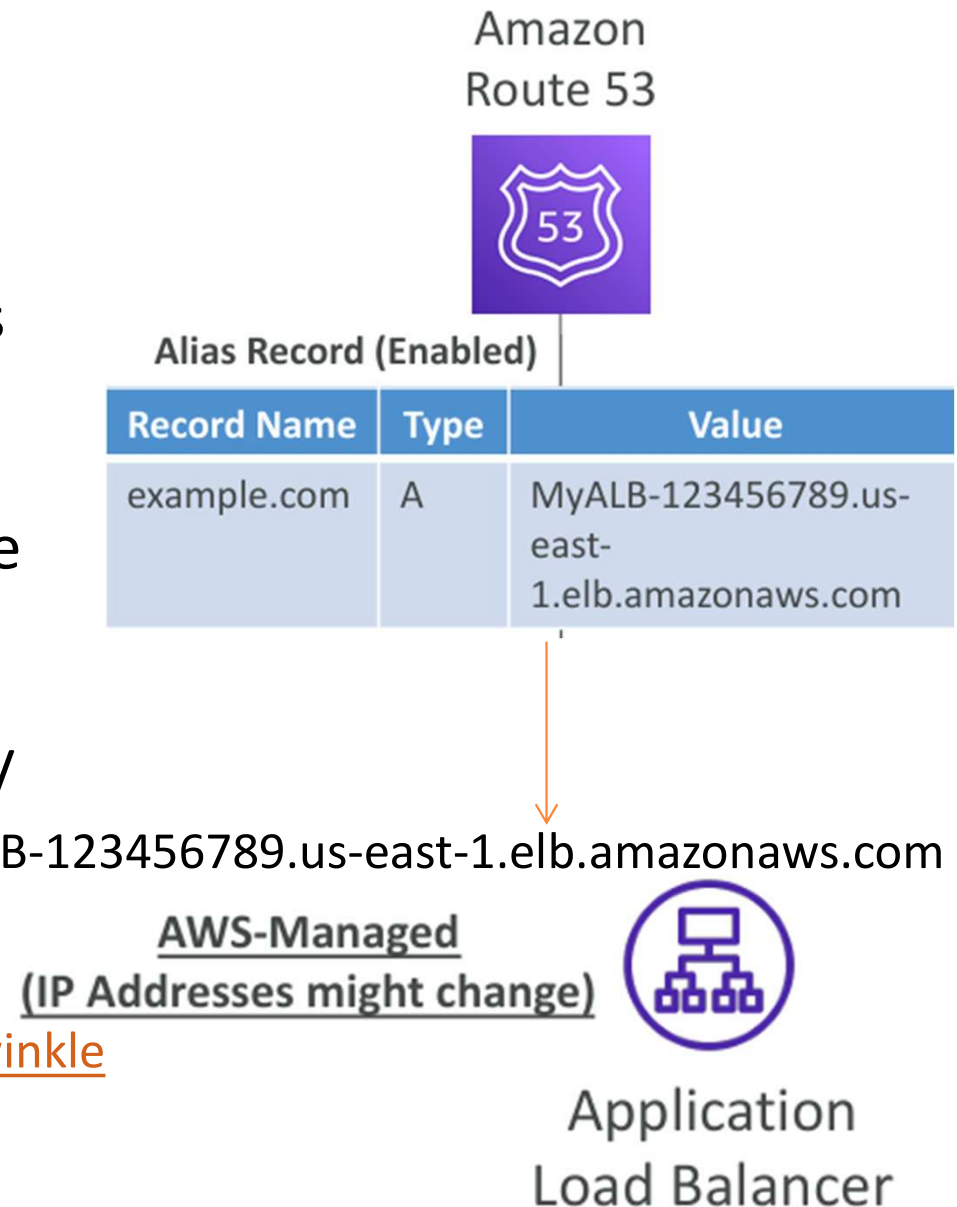
- Points a hostname to an AWS Resource (app.mydomain.com => blabla.amazonaws.com)
- Works for ROOT DOMAIN and NON ROOT DOMAIN (aka mydomain.com)
- Free of charge
- Native health check

<https://www.youtube.com/@techwithtwinkle>

Route 53 – Alias Records

- ▶ Maps a hostname to an AWS resource • An extension to DNS functionality
- ▶ Automatically recognizes changes in the resource's IP addresses
- ▶ Unlike CNAME, it can be used for the top node of a DNS namespace (Zone Apex), e.g.: example.com
- ▶ Alias Record is always of type A/AAAA for AWS resources (IPv4 / IPv6)
- ▶ You can't set the TTL

<https://www.youtube.com/@techwithtwinkle>



Route 53 – Alias Records Targets

- ▶ Elastic Load Balancers
- ▶ CloudFront Distributions
- ▶ API Gateway
- ▶ Elastic Beanstalk environments
- ▶ S3 Websites
- ▶ VPC Interface Endpoints
- ▶ Global Accelerator accelerator
- ▶ Route 53 record in the same hosted zone
- ▶ **You cannot set an ALIAS record for an EC2 DNS name**



Elastic
Load Balancer



Amazon
CloudFront



Amazon
API Gateway



Elastic Beanstalk



S3 Websites



VPC Interface
Endpoints



Global Accelerator



Route 53 Record
(same Hosted Zone)

<https://www.youtube.com/@techwithtwinkle>

Route 53 – Alias Records Targets Hands On

Click on create record

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ **IP-based routing**

- CIDR collections

▼ **Traffic flow**

- Traffic policies
- Policy records

▼ **Domains**

- Registered domains
- Requests

▼ **Resolver**

demo.techtwinklesharma.online was successfully updated.
Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Click the "View status" button to check propagation status.

Records (5) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type ▼ Routing p... ▼ Alias ▼

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-62.org. aws
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
☐	www.techtwinklesharma.online	A	Simple	-	No	172.31.43.104

Buttons: Delete record, Import zone file, Create record

<https://www.youtube.com/@techwithtwinkle>

Create record [Info](#)

Quick create record

Record 1

Record name [Info](#)

myapp

.techtwinklesharma.online

Keep blank to create a record for the root domain.

Alias

Value [Info](#)

www.example.com

Enter multiple values on separate lines.

Record type [Info](#)

CNAME – Routes traffic to another domain name and to some AWS resources

[Switch to wizard](#)

Delete

Enter record name

Select CNAME here

Enter here your load balancer DNS Name

<https://www.youtube.com/@techwithtwinkle>

Value [Info](#)

DemoRoute53ALB-1908756402.ap-south-1.elb.amazonaws.com

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

300

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Simple routing

1m 1h 1d

Add another record

Click on create records

Create records

View existing records

The following table lists the existing records in techtwinklesharma.online.

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ IP-based routing

- CIDR collections

▼ Traffic flow

- Traffic policies
- Policy records

▼ Domains

- Registered domains
- Requests

▼ Resolver

Records (6) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

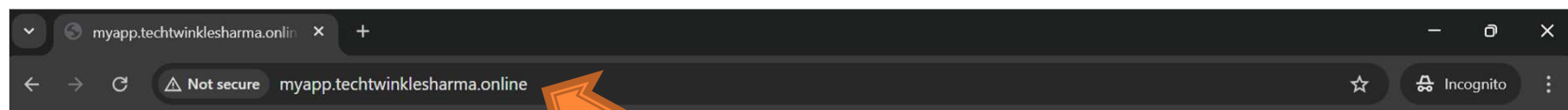
Type Routing p... Alias

1

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	NS	Simple	-	No	ns-1521.awsdns-62.org. ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk.
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-62.org. a
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	myapp.techtwinklesharma.o	CNAME	Simple	-	No	DemoRoute53ALB-19087
☐	test.techtwinklesharma.onlin	A	Simple	-	No	11.22.33.44
☐	www.techtwinklesharma.online		Simple	-	No	172.31.43.104

Record created

Copy this & paste it in chrome browser



Hello world from ip-172-31-47-141.ap-south-1.compute.internal in AZ ap-south-1a

We are accessing our (ALB) load balancer through this URL

<https://www.youtube.com/@techwithtwinkle>

Create Another Record

≡ [Route 53](#) > [Hosted zones](#) > [techwinklesharma.online](#) > Create record



Create record [Info](#)

Quick create record

▼ Record 1

Enter record name

Record name [Info](#)

myalias .techwinklesharma.online

Keep blank to create a record for the root domain.

☐ Alias

On the Alias

Value [Info](#)

192.0.2.235

Enter multiple values on separate lines.

TTI (seconds) [Info](#)

Select Record Type

[Switch to wizard](#)

Delete

Record type [Info](#)

A – Routes traffic to an IPv4 address and some AWS resources

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techtwinklesharma.online > Create record

▼ Record 1 Delete

Record name Info myalias .techtwinklesharma.online

Record type Info A – Routes traffic to an IPv4 address and some AWS resources

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to Info

Alias to Application and Classic Load Balancer

Asia Pacific (Mumbai)

Q dualstack.DemoRoute53ALB-1908756402.ap-south-1.elb.amazonaws.com X

Use: "dualstack.DemoRoute53ALB-1908756402.ap-south-1.elb.amazonaws.com"

Select this **Select ALB Location** **Select ALB**

dualstack.DemoRoute53ALB-1908756402.ap-south-1.elb.amazonaws.com

yes

aws Search [Alt+S] Global ▼ aws-twinkle-v4 (2355-6299-1793) ▼ twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Route traffic to Info

Alias to Application and Classic Load Balancer

Asia Pacific (Mumbai)

Q dualstack.DemoRoute53ALB-1908756402.ap-south-1.elb.amazonaws.com X

Alias hosted zone ID: ZP97RAFLXTNZK

Routing policy Info Simple routing

Evaluate target health ☒ Yes

Add another record

Click on create record  Create records

► View existing records

The following table lists the existing records in techtwinklesharma.online.

<https://www.youtube.com/@techwithtwinkle>

aws twinkl... twinkl

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ IP-based routing

- CIDR collections

▼ Traffic flow

- Traffic policies
- Policy records

▼ Domains

- Registered domains
- Requests

▼ Resolver

Copy & paste it to chrome browser

Record is created

Record for techtwinklesharma.online was successfully created. Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.

Records (7) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-62.or
mo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
myalias.techtwinklesharma.online	A	Simple	-	Yes	dualstack.demoroute5
myapp.techtwinklesharma.online	CNAME	Simple	-	No	DemoRoute53ALB-190
test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
www.techtwinklesharma.online	A	Simple	-	No	172.31.43.104

myalias.techtwinklesharma.onli

Not secure myalias.techtwinklesharma.online

Incognito

Hello world from ip-172-31-47-141.ap-south-1.compute.internal in AZ ap-south-1a

It is showing this

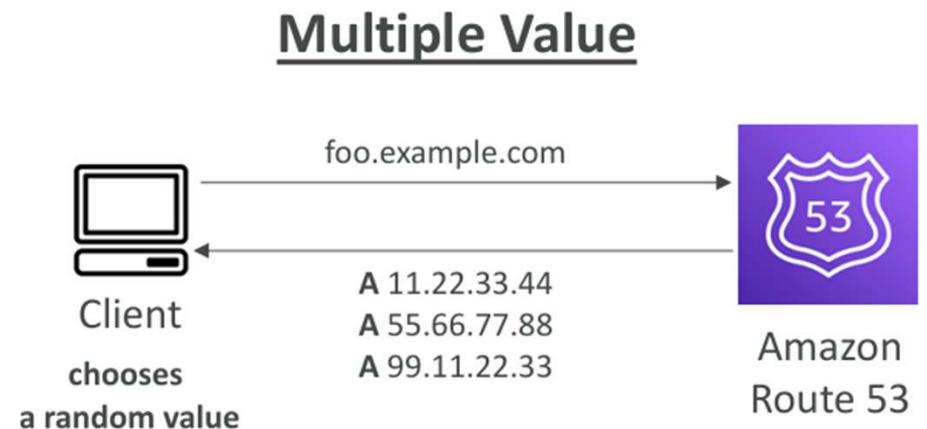
<https://www.youtube.com/@techwithtwinkle>

Route 53 – Routing Policies

- ▶ Define how Route 53 responds to DNS queries
- ▶ Don't get confused by the word "Routing"
 - It's not the same as Load balancer routing which routes the traffic
 - DNS does not route any traffic, it only responds to the DNS queries
- ▶ Route 53 Supports the following Routing Policies
 - Simple
 - Weighted
 - Failover
 - Latency based
 - Geolocation
 - Multi-Value Answer
 - Geoproximity (using Route 53 Traffic Flow feature)

Routing Policies – Simple

- ▶ Typically, route traffic to a single resource
- ▶ Can specify multiple values in the same record
- ▶ **If multiple values are returned, a random one is chosen by the client**
- ▶ When Alias enabled, specify only one AWS resource
- ▶ Can't be associated with Health Checks



Routing Policies – Simple Hands On

Route 53 > Hosted zones > techtwinklesharma.online

Public **techtwinklesharma.online** Info

Delete zone Test record Configure query logging

Hosted zone details Edit hosted zone

Records (7) DNSSEC signing Hosted zone tags (0)

Records (7) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias

1

Create record

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	NS	Simple	-	No	ns-1521.awsdns-62.or ns-918.awsdns-50.net ns-489.awsdns-61.con nc-1867 awednc-11 co

<https://www.youtube.com/@techwithtwinkle>

Create record [Info](#)

Quick create record

[Switch to wizard](#)

Record 1

Record name [Info](#)

simple .techtwinklesharma.online

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

56.155.98.247

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

Record type [Info](#)

A – Routes traffic to an IPv4 address and some AWS resources

Routing policy [Info](#)

Select A

Enter name

Enter IP of any instance

Delete

Value [Info](#)

56.155.98.247

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

20

Recommended values: 60 to 172800 (two days)

1m

1h

1d

Routing policy [Info](#)

Simple routing

Simple routing ✓

Weighted

Simple routing

Geolocation

Latency

Failover

Multivalue answer

IP-based

Geoproximity

Set TTL 20 sec

Select simple routing

Than click on create record

► View existing records

The following table lists the existing records in techtwinklesharma.online.

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ IP-based routing

- CIDR collections

▼ Traffic flow

- Traffic policies
- Policy records

▼ Domains

- Registered domains
- Requests

▼ Resolver

Records (8) [Info](#)

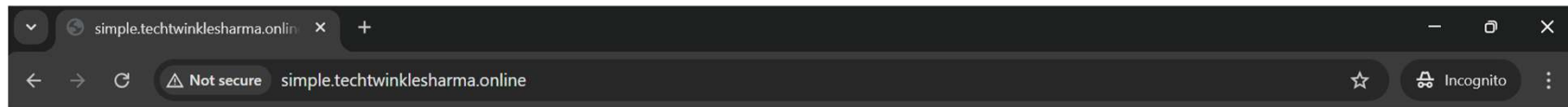
Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type ▼ Routing p... ▼ Alias ▼

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffi
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-62
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	myalias.techtwinklesharma.online	A	Simple	-	Yes	dualstack.demorou
☐	myapp.techtwinklesharma.online	CNAME	Simple	-	No	DemoRoute53ALB-
☐	simple.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
☐	www.techtwinklesharma.online	A	Simple	-	No	172.31.43.104

Copy & paste this url to chrome browser



Hello world from ip-172-31-36-102.ap-northeast-3.compute.internal in AZ ap-northeast-3a



It is showing this

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- ▼ IP-based routing
 - CIDR collections
- ▼ Traffic flow
 - Traffic policies
 - Policy records
- ▼ Domains
 - Registered domains
 - Requests
- ▼ Resolver

Records (1/8) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by pri Type Routing p... Alias

1

	Record name	Type	Routin...
☐	techtwinklesharma.online	SOA	Simple
☐	demo.techtwinklesharma.online	A	Simple
☐	myalias.techtwinklesharma.online	A	Simple
☐	myapp.techtwinklesharma.online	CNAME	Simple
☒	simple.techtwinklesharma.online	A	Simple

Record details

[Edit record](#) **Click on Edit**

Record name
simple.techtwinklesharma.online

Record type
A

Value
56.155.98.247

Alias
No

TTL (seconds)
20

Routing policy
Simple

select

aws Search [Alt+S] Global aws-twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- ▼ IP-based routing
 - CIDR collections
- ▼ Traffic flow
 - Traffic policies
 - Policy records
- ▼ Domains
 - Registered domains
 - Requests
- ▼ Resolver

Records (1/8) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by pri Type Routing p... Alias

1

	Record name	Type	Routin...
☐	techtwinklesharma.online	SOA	Simple
☐	demo.techtwinklesharma.online	A	Simple
☐	myalias.techtwinklesharma.online	A	Simple
☐	myapp.techtwinklesharma.online	CNAME	Simple
☒	simple.techtwinklesharma.online	A	Simple

Add another IP of another instance & click on save

Record details

Record name
simple.techtwinklesharma.online

Record type
A – Routes traffic to an IPv4 address and s...

Alias
☐ Alias

Value
56.155.98.247
98.130.57.108

Enter multiple values on separate lines.

TTL (seconds)
20 1m 1h 1d
Recommended values: 60 to 172800 (two days)

Routing policy
Simple

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ IP-based routing

- CIDR collections

▼ Traffic flow

- Traffic policies
- Policy records

▼ Domains

- Registered domains
- Requests

▼ Resolver

Records (8) [Info](#)

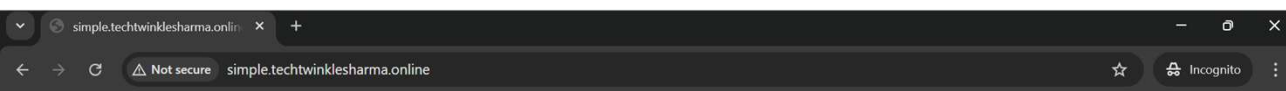
Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type ▼ Routing p... ▼ Alias ▼

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffi
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	myalias.techtwinklesharma.online	A	Simple	-	Yes	dualstack.demorou
☐	myapp.techtwinklesharma.online	CNAME	Simple	-	No	DemoRoute53ALB-
☐	simple.techtwinklesharma.online	A	Simple	-	No	56.155.98.247 98.130.57.108
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
☐	www.techtwinklesharma.online	A	Simple	-	No	172.31.43.104

2 records it I showing



Hello world from ip-172-31-36-102.ap-northeast-3.compute.internal in AZ ap-northeast-3a

Hello world from ip-172-31-36-177.ap-south-2.compute.internal in AZ ap-south-2b

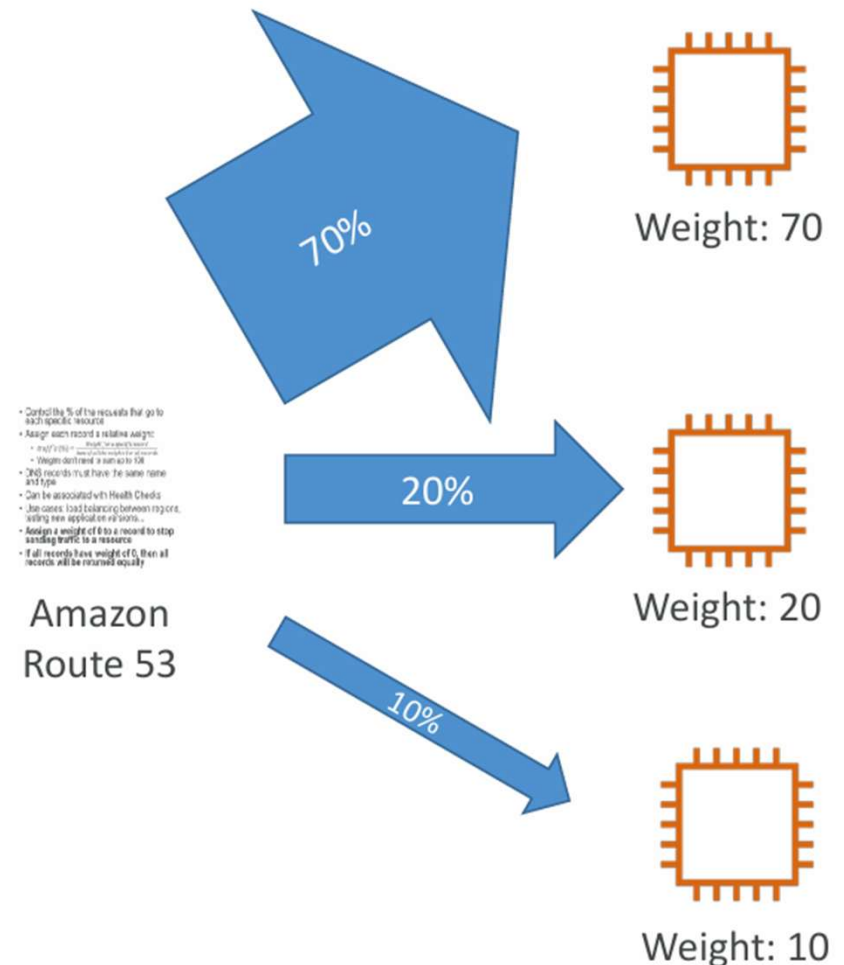


After 20 seconds it will show this

<https://www.youtube.com/@techwithtwinkle>

Routing Policies – Weighted

- ▶ Control the % of the requests that go to each specific resource
- ▶ Assign each record a relative weight:
- ▶
$$traffic\ (%) = \frac{Weight\ for\ a\ specific\ record}{Sum\ of\ all\ the\ weights\ for\ all\ records}$$
- ▶ DNS records must have the same name and type
- ▶ Can be associated with Health Checks
- ▶ Use cases: load balancing between regions, testing new application versions...
- ▶ Assign a weight of 0 to a record to stop sending traffic to a resource
- ▶ If all records have weight of 0, then all records will be returned equally



Routing Policies – Weighted Hands On

Route 53 > Hosted zones > techtwinklesharma.online

Click on create record

Create record

Records (8) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traff
☐	techtwinklesharma.online	NS	Simple	-	No	ns-1521.awsdns-6 ns-918.awsdns-50 ns-489.awsdns-61 ns-1867.awsdns-4
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-6
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	myalias.techtwinklesharma.online	A	Simple	-	Yes	dualstack.demorou
☐	myapp.techtwinklesharma.online	CNAME	Simple	-	No	DemoRoute53ALB
☐	simple.techtwinklesharma.online	A	Simple	-	No	56.155.98.247 98.130.57.108
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44

<https://www.youtube.com/@techwithtwinkle>

Create record [Info](#)

Quick create record

[Switch to wizard](#)

Record 1

Record name [Info](#)

weighted .techwinklesharma.online

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

56.155.98.247

Enter multiple values on separate lines.

Record type [Info](#)

A – Routes traffic to an IPv4 address and some AWS resources

Delete

Enter record name

Select A record

Copy & paste IP of instance

<https://www.youtube.com/@techwithtwinkle>

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

3

Recommended values: 60 to 172800 (two days)

1m

1h

1d

TTL 3 sec

Routing policy [Info](#)

Weighted

Select weighted

Weight

The weight of each record that has the same name and type determines the proportion of DNS queries that Route 53 will respond to using that record.

10

The weight can be a number between 0 and 255. If you specify 0, Route 53 stops responding to DNS queries using this record.

Health check ID - optional [Info](#)

Choose health check

Enter 10

Record ID [Info](#)

northeast

Enter location of that instance

Add another record

Recommended values: 60 to 172800 (two days)

Weight

The weight of each record that has the same name and type determines the proportion of DNS queries that Route 53 will respond to using that record.

10

The weight can be a number between 0 and 255. If you specify 0, Route 53 stops responding to DNS queries using this record.

Record ID

Info

northeast

Health check ID - optional

Info

Choose health check



Click on Add another record



Add another record

Cancel

Create records

View existing records

The following table lists the existing records in techwinklesharma.online.

northeast

Record 2

Enter Record name

Record name

Info

weighted

.techwinklesharma.online

Keep blank to create a record for the root domain.

Alias

Value

Info

98.130.57.108

Enter record type

Record type

Info

A – Routes traffic to an IPv4 address and some AWS resources

Delete

Enter IP of another instance

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techwinklesharma.online > Create record

Enter multiple values on separate lines.

TTL (seconds) Info

3

Recommended values: 60 to 172800 (two days)

Select TTL 3

Routing policy Info

Weighted

Select weighted routing policy

Weight

The weight of each record that has the same name and type determines the proportion of DNS queries that Route 53 will respond to using that record.

70

The weight can be a number between 0 and 255. If you specify 0, Route 53 stops responding to DNS queries using this record.

Health check ID - optional Info

Choose health check

Record ID Info

south

Enter location of instance

After that click on add another record

Add another record

Route 53 > Hosted zones > techwinklesharma.online > Create record

▼ Record 3

Record name Info

weighted

Keep blank to create a record for the root domain.

Record type Info

A – Routes traffic to an IPv4 address and some AWS resources

Enter Record A

Enter record name

Enter another IP of another instance

Value Info

13.127.17.56

https://www.youtube.com/@techwithtwinkle

Enter multiple values on separate lines.

TTL (seconds) Info

3

Recommended values: 60 to 172800 (two days)

Enter TTL 3

Routing policy Info

Weighted

Select weighted policy

Alias

Info

Delete

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

3

Recommended values: 60 to 172800 (two days)

1m

1h

1d

Routing policy [Info](#)

Weighted

Weight

The weight of each record that has the same name and type determines the proportion of DNS queries that Route 53 will respond to using that record.

20

The weight can be a number between 0 and 255. If you specify 0, Route 53 stops responding to DNS queries using this record.

Health check ID - optional [Info](#)

Choose health check

Record ID [Info](#)

south

Enter 20

Enter location of instance

Add another record

Weight

The weight of each record that has the same name and type determines the proportion of DNS queries that Route 53 will respond to using that record.

20

The weight can be a number between 0 and 255. If you specify 0, Route 53 stops responding to DNS queries using this record.

Record ID [Info](#)

south

Health check ID - optional [Info](#)

Choose health check

Add another record

Click on create records

Create records

View existing records

The following table lists the existing records in techtwinklesharma.online.

<https://www.youtube.com/@techwithtwinkle>

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)

▼ IP-based routing

- CIDR collections

▼ Traffic flow

- Traffic policies
- Policy records

▼ Domains

- Registered domains
- Requests

▼ Resolver

Records (11) [Info](#)

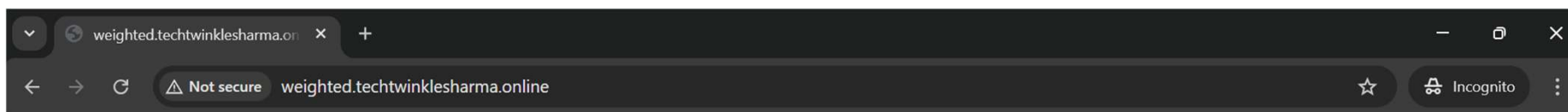
Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type ▼ Routing p... ▼ Alias ▼

☐	Record ... ▼	Type ▼	Routin... ▼	Differ... ▼	Alias ▼	Value/Route traffic to ▼	TTL (s... ▼
☐	myapp.tec...	CNAME	Simple	-	No	DEMOROUTES3AED-1300730...	300
☐	simple.tec...	A	Simple	-	No	56.155.98.247 98.130.57.108	20
☐	test.techt...	A	Simple	-	No	11.22.33.44	300
☐	weighted...	A	Weighted	10	No	56.155.98.247	3
☐	weighted...	A	Weighted	70	No	98.130.57.108	3
☐	weighted...	A	Weighted	20	No	13.127.17.56	3
☐	www.tech...	A	Simple	-	No	172.31.43.104	300

3 weighted records created



Hello world from ip-172-31-36-177.ap-south-2.compute.internal in AZ ap-south-2b

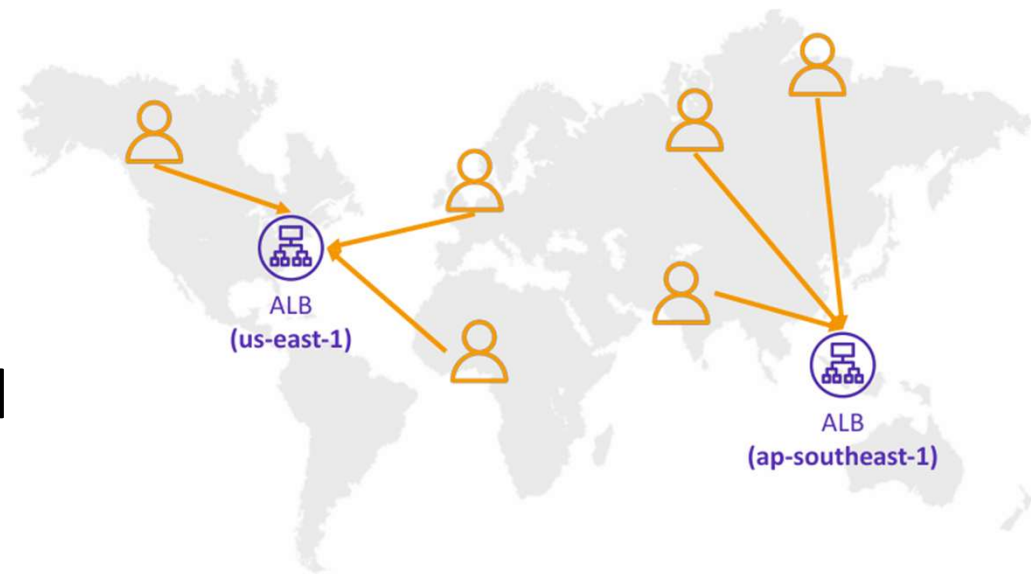


It is showing this After few seconds it will change on multiple refresh

<https://www.youtube.com/@techwithtwinkle>

Routing Policies – Latency-based

- ▶ Redirect to the resource that has the least latency close to us
- ▶ Super helpful when latency for users is a priority
- ▶ **Latency is based on traffic between users and AWS Regions**
- ▶ Germany users may be directed to the US (if that's the lowest latency)
- ▶ Can be associated with Health Checks (has a failover capability)



Route 53 > Hosted zones > techtwinklesharma.online

Click on create record

Records (11) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias < 1 > ⚙

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traf
☐	techtwinklesharma.online	NS	Simple	-	No	ns-1521.awsdns-... ns-918.awsdns-50... ns-489.awsdns-6... ns-1867.awsdns-...
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-...
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	myalias.techtwinklesharma.online	A	Simple	-	Yes	dualstack.demorc
☐	myapp.techtwinklesharma.online	CNAME	Simple	-	No	DemoRoute53ALI
☐	simple.techtwinklesharma.online	A	Simple	-	No	56.155.98.247 98.130.57.108
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Create record Info

Quick create record

Record 1

Record name Info

latency

.techtwinklesharma.online

Keep blank to create a record for the root domain.

☐ Alias

Value Info

13.127.17.56

Enter multiple values on separate lines.

TTL (seconds) Info

Record type Info

A – Routes traffic to an IPv4 address and some AWS resources

Select record A

[Switch to wizard](#)

Delete

Enter IP Address of instance

<https://www.youtube.com/@techwithtwinkle>

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Value

Info

13.127.17.56

Enter multiple values on separate lines.

TTL (seconds)

Info

300

1m

1h

1d

Recommended values: 60 to 172800 (two days)

Routing policy

Info

Latency

Region

The Amazon EC2 region where the resource that you specified in this record resides. You can only create one latency record for each Amazon EC2 region. You aren't required to create latency records for all Amazon EC2 regions.

Asia Pacific (Mumbai)

Health check ID - optional

Info

Choose health check

Record ID

Info

ap-south-1

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Recommended values: 60 to 172800 (two days)

Region

The Amazon EC2 region where the resource that you specified in this record resides. You can only create one latency record for each Amazon EC2 region. You aren't required to create latency records for all Amazon EC2 regions.

Asia Pacific (Mumbai)

Record ID

Info

ap-south-1

Health check ID - optional

Info

Choose health check

Click on Another record

Add another record

Cancel

Create records

View existing records

The following table lists the existing records in techtwinklesharma.online.

https://www.youtube.com/@techwithtwinkle

aws twinkl v4 (2355-6299-1793) twinkl

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Record 2

Record name [Info](#)

latency .techtwinklesharma.online

Keep blank to create a record for the root domain.

Alias

Value [Info](#)

98.130.57.108

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

300

Recommended values: 60 to 172800 (two days)

1m 1h 1d

Record type [Info](#)

A - Routes traffic to an IPv4 address and some AWS resources

Enter record name

Enter record A

Paste IP of another instance

Select latency

<https://www.youtube.com/@techwithtwinkle>

aws twinkl v4 (2355-6299-1793) twinkl

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

300

Recommended values: 60 to 172800 (two days)

1m 1h 1d

Routing policy [Info](#)

Latency

Region

The Amazon EC2 region where the resource that you specified in this record resides. You can only create one latency record for each Amazon EC2 region. You aren't required to create latency records for all Amazon EC2 regions.

Asia Pacific (Hyderabad)

Record ID [Info](#)

ap-south-2

Health check ID - optional [Info](#)

Choose health check

Select location of instance

Enter location

After all this click on another record

Add another record

aws twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

ap-south-2

▼ Record 3

Enter record name

Record name latency .techtwinklesharma.online

Record type A – Routes traffic to an IPv4 address and some AWS resources

Select record type

Alias

Select IP of 3rd instance

Value 56.155.98.247

<https://www.youtube.com/@techwithtwinkle>

TTL (seconds) 300

Routing policy Latency

aws twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

TTL (seconds) 300

Routing policy Latency

Select latency

Health check ID - optional

Select region of instance

Region Asia Pacific (Osaka)

Enter region

Record ID ap-northeast-3

Click on create records

Create records

View existing records

aws twinkl [255-623-1755] Global twinkl

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles [New](#)
- ▼ **IP-based routing**
 - CIDR collections
- ▼ **Traffic flow**
 - Traffic policies
 - Policy records
- ▼ **Domains**
 - Registered domains
 - Requests
- ▼ **Resolver**

Records (14) [Info](#) [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias < 1 > ⚙

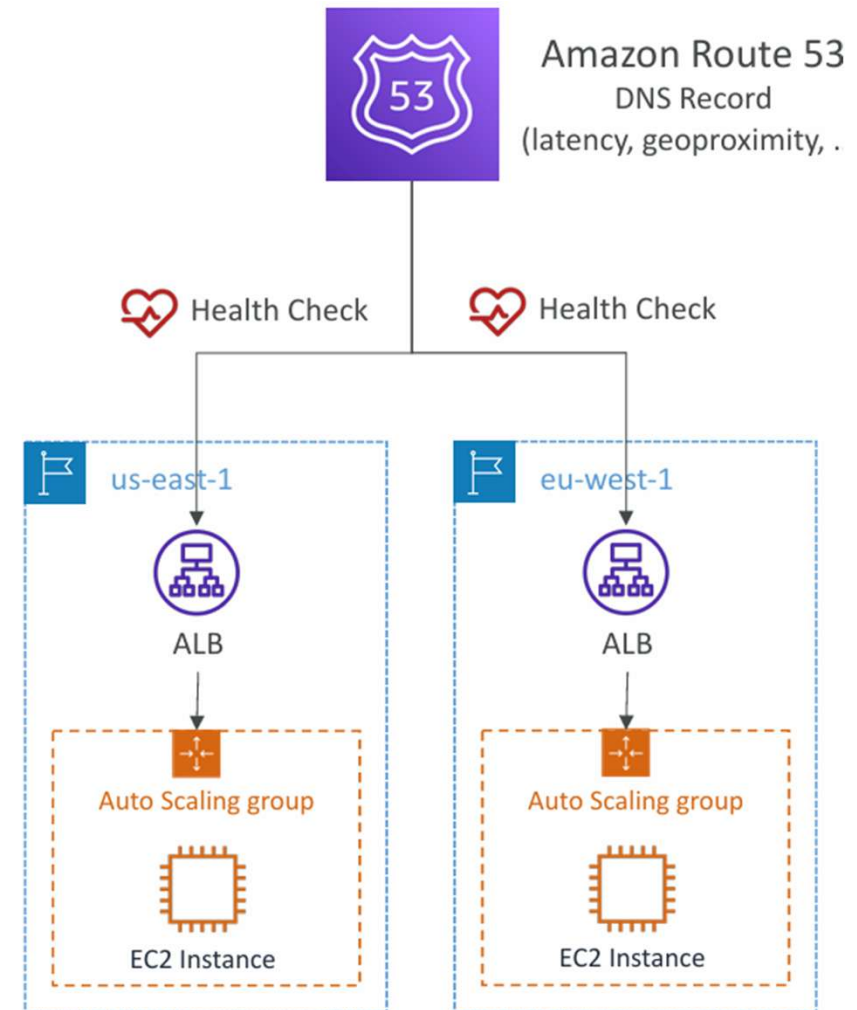
☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	SOA	Simple	-	No	ns-1521.awsdns-62.org.
☐	demo.techtwinklesharma.online	A	Simple	-	No	56.155.98.247
☐	latency.techtwinklesharma.online	A	Latency	Asia Paci...	No	56.155.98.247
☐	latency.techtwinklesharma.online	A	Latency	Asia Paci...	No	13.127.17.56
☐	latency.techtwinklesharma.online	A	Latency	Asia Paci...	No	98.130.57.108
☐	myalias.techtwinklesharma.online	A	Simple	-	Yes	dualstack.demoroute53a
☐	myapp.techtwinklesharma.online	CNAME	Simple	-	No	DemoRoute53ALB-1908
☐	simple.techtwinklesharma.online	A	Simple	-	No	56.155.98.247 98.130.57.108
☐	test.techtwinklesharma.online	A	Simple	-	No	11.22.33.44
☐	weighted-techtwinklesharma-online	A	Weighted	10	No	56.155.98.247

Latency created

<https://www.youtube.com/@techwithtwinkle>

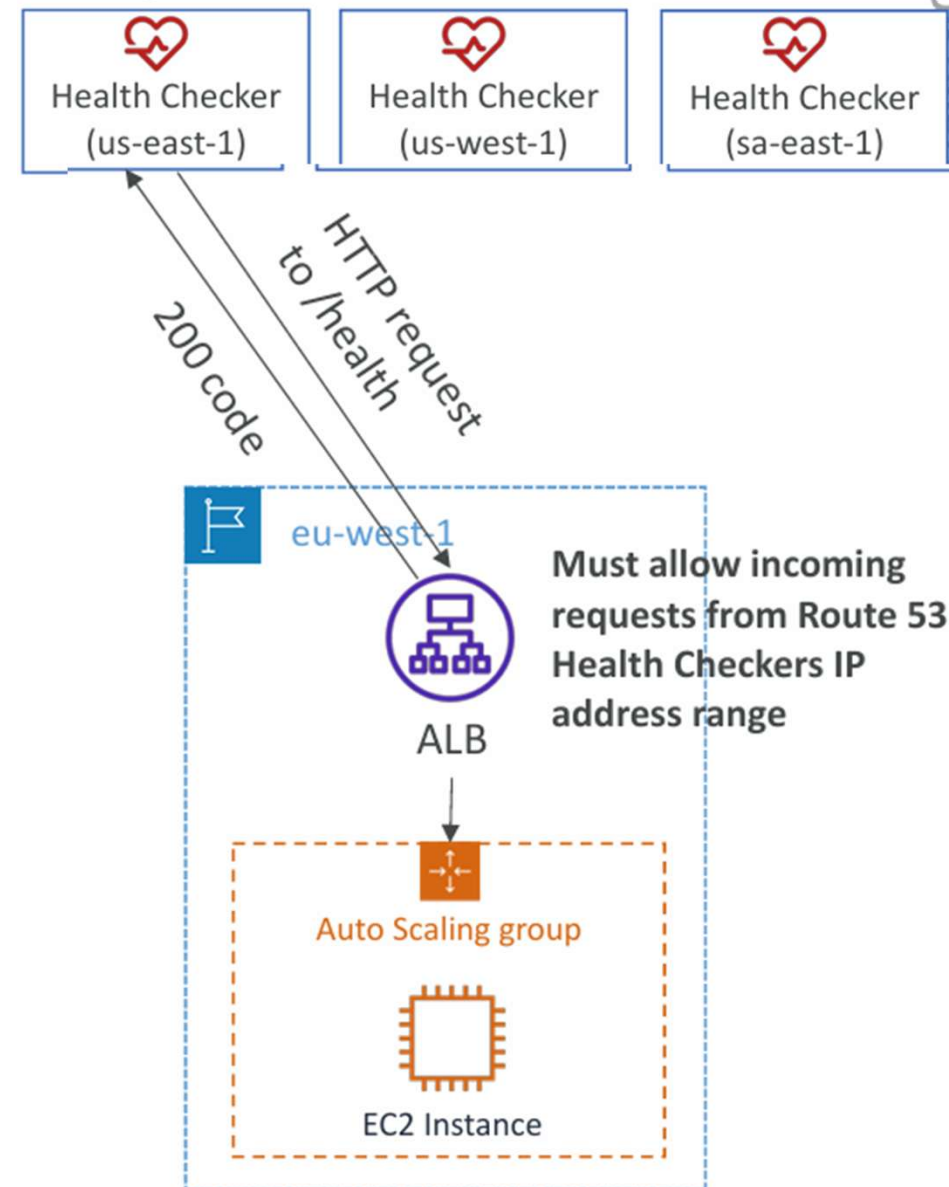
Route 53 – Health Checks

- ▶ HTTP Health Checks are only for **public resources**
 - ▶ Health Check => Automated DNS Failover:
 1. Health checks that monitor an endpoint (application, server, other AWS resource)
 2. Health checks that monitor other health checks (Calculated Health Checks)
 3. Health checks that monitor CloudWatch Alarms (full control !!) – e.g., throttles of DynamoDB, alarms on RDS, custom metrics, ... (helpful for private resources)
- Health Checks are integrated with CW metrics



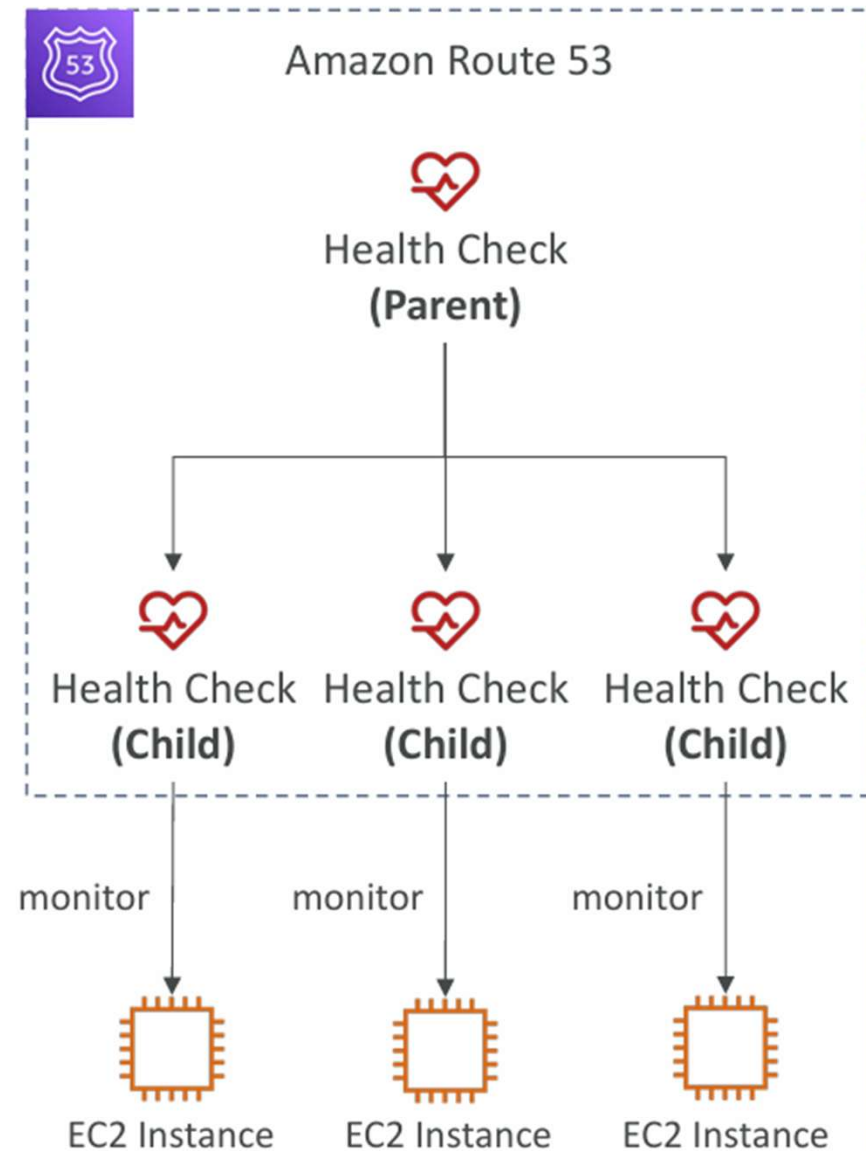
Health Checks – Monitor an Endpoint

- ▶ About 15 global health checkers will check the endpoint health
- Healthy/Unhealthy Threshold – 3 (default)
- Interval – 30 sec (can set to 10 sec – higher cost)
- Supported protocol: HTTP, HTTPS and TCP
- If > 18% of health checkers report the endpoint is healthy, Route 53 considers it Healthy. Otherwise, it's Unhealthy
- Ability to choose which locations you want Route 53 to use
- ▶ Health Checks pass only when the endpoint responds with the 2xx and 3xx status codes
- ▶ Health Checks can be setup to pass / fail based on the text in the first 5120 bytes of the response
- ▶ Configure your router/firewall to allow incoming requests from Route 53 Health Checkers



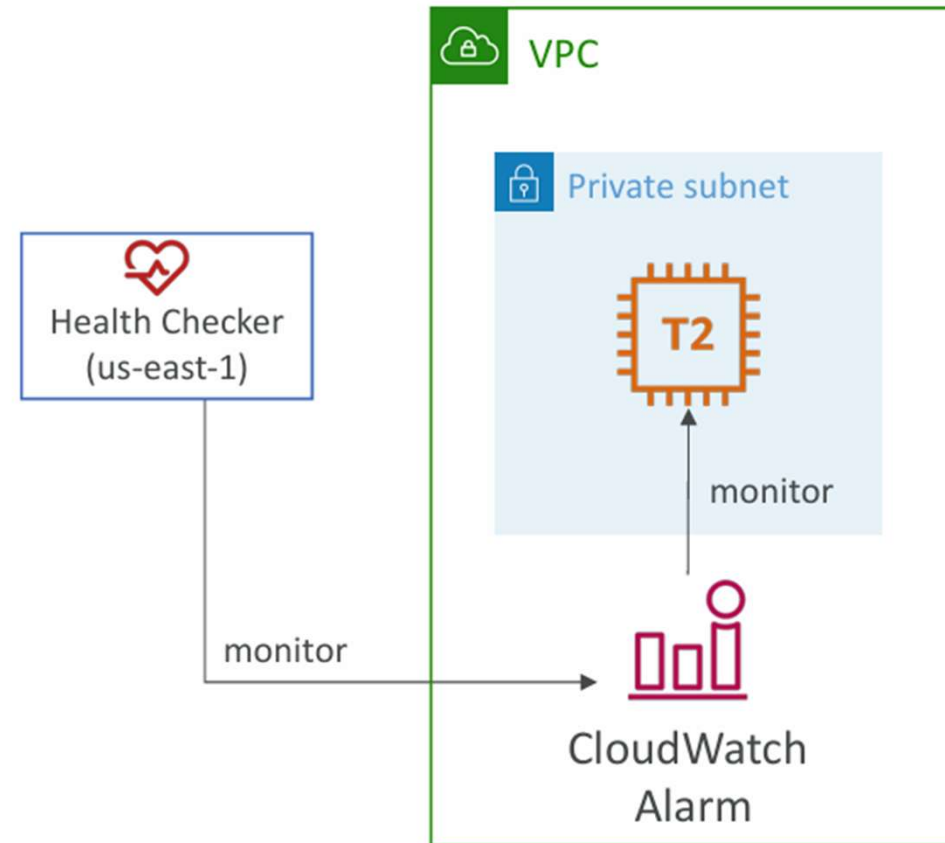
Route 53 – Calculated Health Checks

- ▶ Combine the results of multiple Health Checks into a single Health Check
- ▶ You can use **OR, AND, or NOT**
- ▶ Can monitor up to 256 Child Health Checks
- ▶ Specify how many of the health checks need to pass to make the parent pass
- ▶ Usage: perform maintenance to your website without causing all health checks to fail

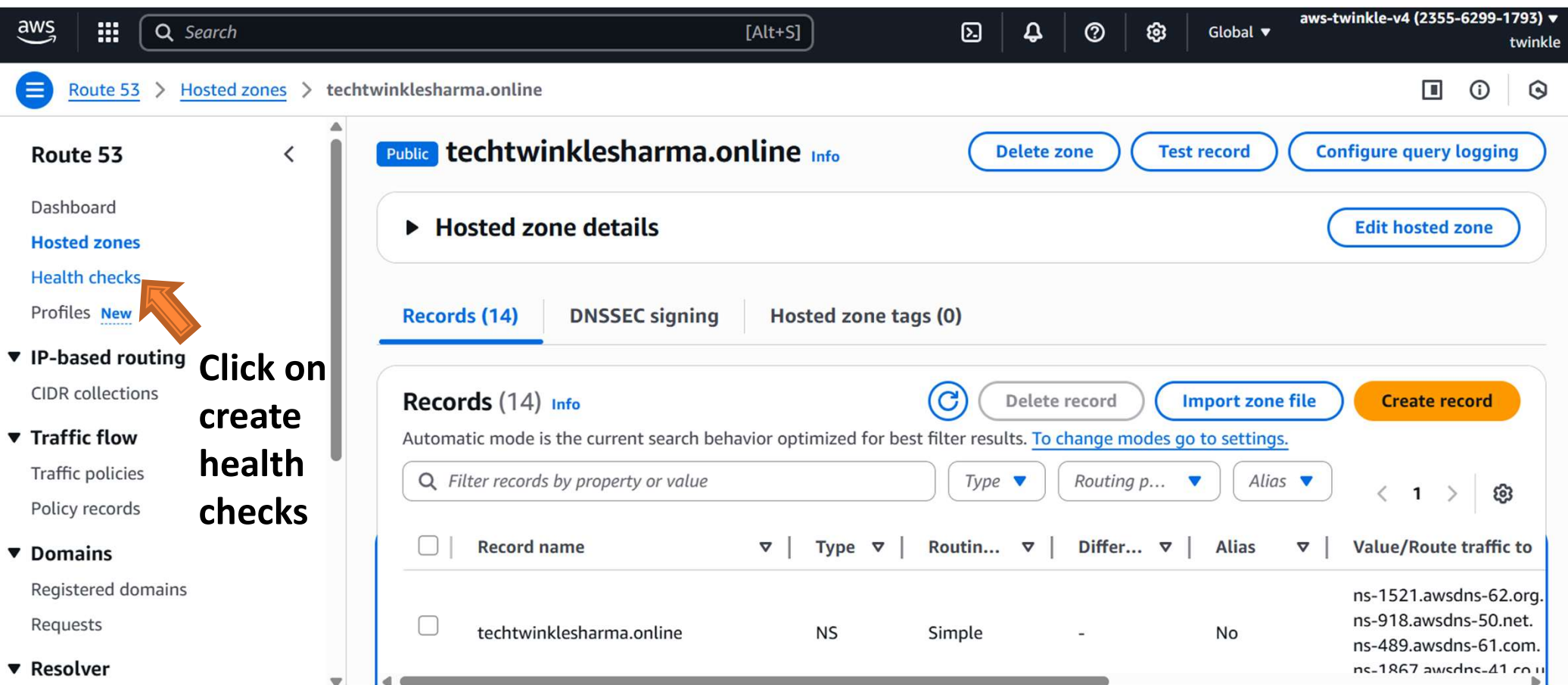


Health Checks – Private Hosted Zones

- ▶ Route 53 health checkers are outside the VPC
- ▶ They can't access private endpoints (private VPC or on-premises resource)
- ▶ You can create a CloudWatch Metric and associate a CloudWatch Alarm, then create a Health Check that checks the alarm itself



Route 53 – Health Checks Hands On



The screenshot shows the AWS Route 53 console interface. On the left sidebar, the 'Health checks' link is highlighted with an orange arrow. The main content area shows the 'Hosted zone details' for 'techtwinklesharma.online'. The 'Records (14)' tab is selected, displaying a table of DNS records. The table has columns for Record name, Type, Routing, Differ..., Alias, and Value/Route traffic to. The first record is 'techtwinklesharma.online' with Type 'NS' and Routing 'Simple'. The Value/Route traffic to column lists several AWS DNS endpoints.

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles **New**
- ▼ **IP-based routing**
 - CIDR collections
- ▼ **Traffic flow**
 - Traffic policies
 - Policy records
- ▼ **Domains**
 - Registered domains
 - Requests
- ▼ **Resolver**

Public techtwinklesharma.online Info

Hosted zone details Edit hosted zone

Records (14) Info DNSSEC signing Hosted zone tags (0)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	techtwinklesharma.online	NS	Simple	-	No	ns-1521.awsdns-62.org. ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk

<https://www.youtube.com/@techwithtwinkle>

Health checks (0) [Info](#) Click on create health checks

Route 53 health checks monitor the health and performance of your application's servers and endpoints.

< 1 > [Settings](#)

☐	ID	Name	Details	Status in last 24 hours	Current s	Actions
--------------------------	----	------	---------	-------------------------	-----------	---------

No health checks to display.

[Create health check](#)

<https://www.youtube.com/@techwithtwinkle>

Select a health check

Create health check [Info](#)

Create a new health check to monitor the health of a specified resource.

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#)

Name - optional

A friendly name that lets you easily find a health check in the dashboard.

ap-south-1

The name must have 1-256 characters, Valid characters; A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Enter name

Resource

☒ Endpoint

Establish a connection with the resource to determine its health status.

☐ Calculated health check

The status of the health check is based on the status of the other health checks.

☐ CloudWatch alarm

The status of the health check is based on the state of a specified CloudWatch alarm.

Specify endpoint by

☒ IP address

☐ Domain name

Select IP

Select Endpoint

aws twinkle

Route 53 > Health checks > Create health check

☐ Domain name

IP address
The path can be any value for which your endpoint will return an HTTP status code of 2xx or 3xx when the endpoint is healthy.

HTTP 13.127.17.56

▼ **Advanced configuration**

Request interval
In this time interval you will receive one request per health checker.

☒ Standard (30 seconds)
☐ Fast (10 seconds)

Failure threshold
The number of consecutive health check failures that is considered unhealthy.

3

☐ String matching - *optional*
Search the response body for a specified search string. The endpoint is considered healthy if the entire value appears within the first 5120 bytes of the response body.



Enter the IP Address

aws twinkle

Route 53 > Health checks > Create health check

☒ US West (Oregon) ☒ Asia Pacific (Singapore)
☒ EU (Ireland) ☒ Asia Pacific (Sydney)
☒ Asia Pacific (Tokyo) ☒ South America (Sao Paulo)

Host name - optional
This value will be passed to the endpoint in a Host header.

example.com

Tags - optional
A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tags

You can add up to 50 more tags.

<https://www.youtube.com/@techwithtwinkle>

Click on create Health Checks



Create health check

Create Second health check

aws   [Alt+S]     Global aws-twinkle-v4 (2355-6299-1793) ▼ twinkle

 [Route 53](#) > [Health checks](#) > Create health check  

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#)

Name - optional
A friendly name that lets you easily find a health check in the dashboard.

 **Enter name**

The name must have 1-256 characters, Valid characters; A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource

☒ **Endpoint**
Establish a connection with the resource to determine its health status.

☐ **Calculated health check**
The status of the health check is based on the status of the other health checks.

☐ **CloudWatch alarm**
The status of the health check is based on the state of a specified CloudWatch alarm.

Specify endpoint by

☒ IP address

☐ Domain name

IP address
The path can be any value for which your endpoint will return HTTP status code of 2xx or 3xx when the endpoint is healthy.

 **Enter IP**

Then click on create health check

<https://www.youtube.com/@techwithtwinkle>

Create Third health check

aws   [Alt+S]     Global  aws-twinkle-v4 (2355-6299-1793)  

 [Route 53](#) > [Health checks](#) > Create health check  

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#)

Name - optional
A friendly name that lets you easily find a health check in the dashboard.

 **Enter name**

The name must have 1-256 characters, Valid characters; A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource

☒ **Endpoint**
Establish a connection with the resource to determine its health status.

☐ **Calculated health check**
The status of the health check is based on the status of the other health checks.

☐ **CloudWatch alarm**
The status of the health check is based on the state of a specified CloudWatch alarm.

Specify endpoint by

☒ IP address

☐ Domain name

IP address
The path can be any value for which your endpoint will return a 2xx or 3xx status code when the endpoint is healthy.

 **Enter IP**

<https://www.youtube.com/@techwithtwinkle>

Then click on create health check

Route 53 > Health checks

Route 53

Dashboard

Hosted zones

Health checks

Profiles New

▼ IP-based routing

CIDR collections

▼ Traffic flow

Traffic policies

Policy records

▼ Domains

Registered domains

Requests

▼ Resolver

https://www.youtube.com/@techwithtwinkle

Health checks (3) Info

Route 53 health checks monitor the health and performance of your application's servers and endpoints.

Find health check

< 1 >

⚙

☐	ID	Name	Details	Status in last 24 hours	Actions
☐	222d366a-d344-40bc-...	ap-northeast-3	http://56....		⋮
☐	2c0911f2-3b19-4d9d-...	ap-south-2	http://98....		⋮
☐	9a283bbb-6337-4a17-...	ap-south-1	http://13....		⋮

Health checks created now

Select a health check

Route 53 > Health checks

Route 53

Dashboard

Hosted zones

Health checks

Profiles New

▼ IP-based routing

CIDR collections

▼ Traffic flow

Traffic policies

Policy records

▼ Domains

Registered domains

Requests

▼ Resolver

Health checks (3) Info

Route 53 health checks monitor the health and performance of your application's servers and endpoints.

Find health check

< 1 >

⚙

Details	Status in last 24 hours	Current s...	Alarm	State	Actions
http://56....		✔ Healthy	None, ...	✔ Enabled	⋮
http://98....		✔ Healthy	None, ...	✔ Enabled	⋮
http://13....		✔ Healthy	None, ...	✔ Enabled	⋮

Healthy health checks

Select a health check

Calculated health check Hands On

aws   [Alt+S]     Global  aws-twinkle-v4 (2355-6299-1793)  twinkle

 [Route 53](#) > [Health checks](#) > Create health check  

Create health check [Info](#)

Create a new health check to monitor the health of a specified resource.

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#).

Name - optional
A friendly name that lets you easily find a health check in the dashboard.

The name must have 1-256 characters, Valid characters; A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource

☐ Endpoint
Establish a connection with the resource to determine its health status.

☒ Calculated health check
The status of the health check is based on the status of the other health checks.

☐ CloudWatch alarm
The status of the health check is based on the state of a specified CloudWatch alarm.

Health checks to monitor

Enter the name of health check

Select this

Click here

<https://www.youtube.com/@techwithtwinkle>

calculated

The name must have 1-256 characters, Valid characters; A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource

☐ Endpoint

Establish a connection with the resource to determine its health status.

☒ Calculated health check

The status of the health check is based on the status of the other health checks.

☐ CloudWatch alarm

The status of the health check is based on the state of a specified CloudWatch alarm.

Health checks to monitor

Search for health checks



- ☒ ap-northeast-3
222d366a-d344-40bc-b24d-323a8a230c65
- ☒ ap-south-2
2c0911f2-3b19-4d9d-af68-2e98a4469b6b
- ☒ ap-south-1
9a283bbb-6337-4a17-8c08-0341291a3ab7



Select these health check



Search

[Alt+S]



Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

222d366a-d344-40bc-b24d-323a8a230c65

2c0911f2-3b19-4d9d-af68-2e98a4469b6b

Show all selected health checks (+1)

Report healthy when

☒ All health checks are healthy

☐ A minimum number of health checks that must be considered healthy

<https://www.youtube.com/@techwithtwinkle>

Tags - optional

A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tags

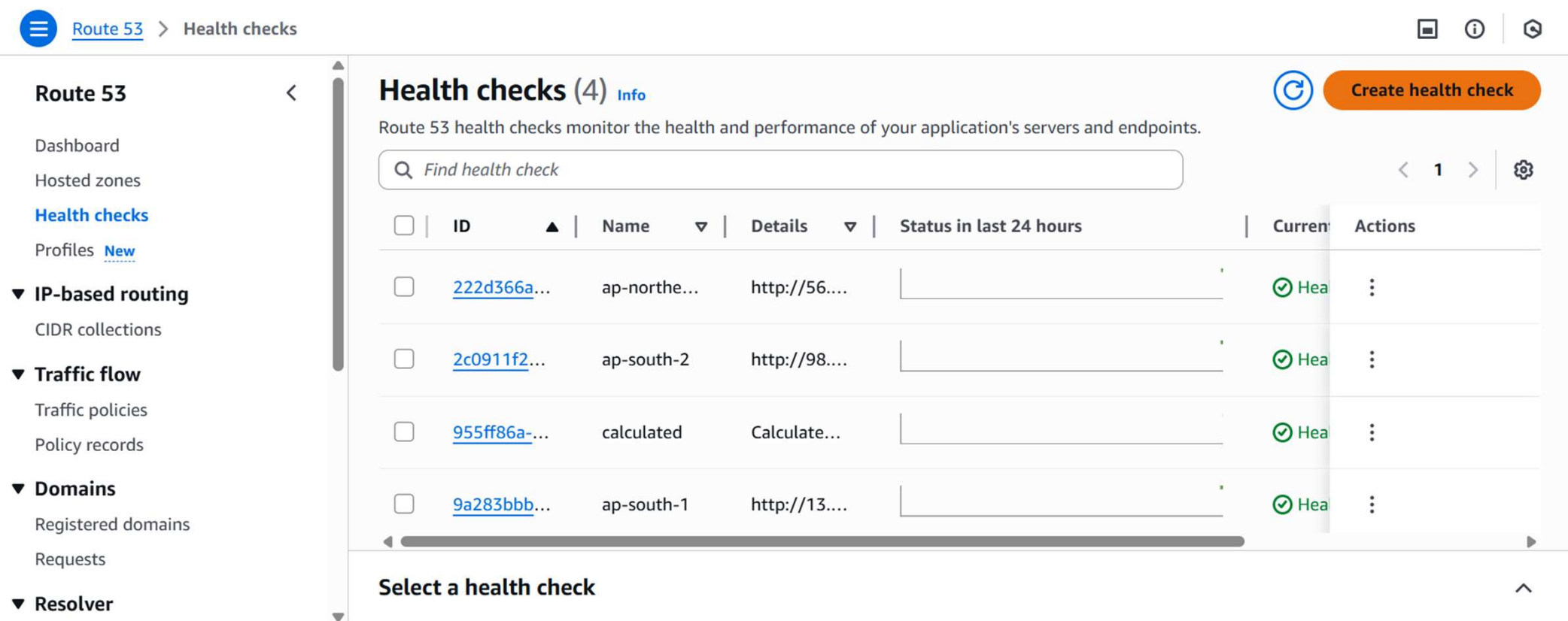
You can add up to 50 more tags.

Click on create Health check



Create health check

All Health checks created



The screenshot displays the AWS Route 53 Health Checks console. The left sidebar shows the navigation menu with 'Route 53' selected. The main content area is titled 'Health checks (4)' and includes a search bar and a table of health checks. The table has columns for selection, ID, Name, Details, Status in last 24 hours, Current status, and Actions. Four health checks are listed, all with a 'Healthy' status.

Route 53 > Health checks

Health checks (4) [Info](#)

Route 53 health checks monitor the health and performance of your application's servers and endpoints.

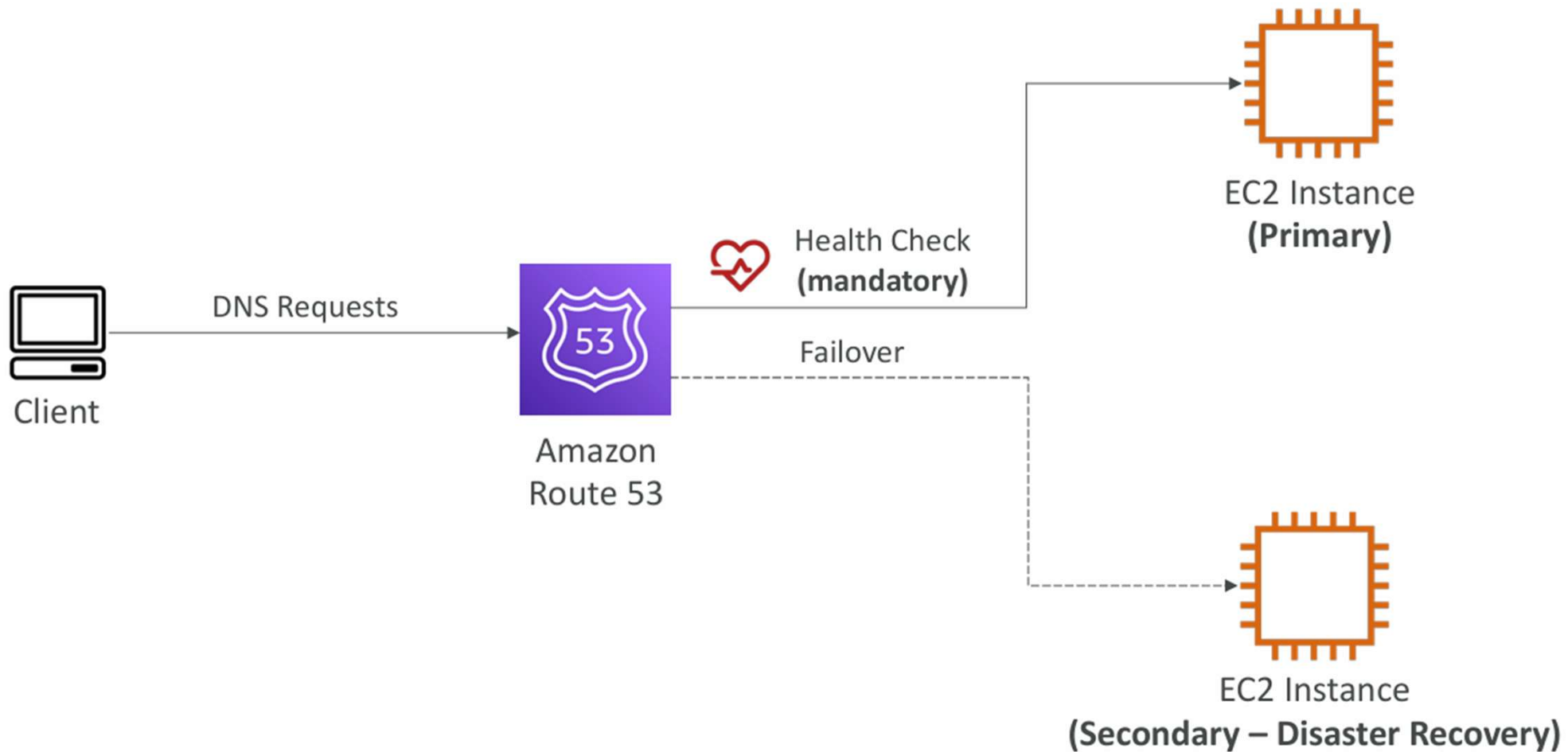
< 1 >

☐	ID	Name	Details	Status in last 24 hours	Current	Actions
☐	222d366a...	ap-northe...	http://56...		✓ Healthy	⋮
☐	2c0911f2...	ap-south-2	http://98...		✓ Healthy	⋮
☐	955ff86a...	calculated	Calculate...		✓ Healthy	⋮
☐	9a283bbb...	ap-south-1	http://13...		✓ Healthy	⋮

Select a health check

<https://www.youtube.com/@techwithtwinkle>

Routing Policies – Failover (Active-Passive)



Routing Policies – Failover (Active-Passive) Hands On

The screenshot shows the AWS Route 53 console interface. The left sidebar contains the navigation menu with 'Route 53' selected. The main content area displays 'Hosted zones (1)' with a table listing the hosted zones. The table has columns: Hosted zone name, Type, Created by, Record co..., Description, and Hosted z... The first row shows a hosted zone named 'techtwinklesharma.online' of type 'Public', created by 'Route 53', with 14 records and no description. The ID is 'Z0985708...'. There are buttons for 'View details', 'Edit', 'Delete', and 'Create hosted zone' at the top right of the table. A search bar is also present above the table.

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles [New](#)
- ▼ **IP-based routing**
 - CIDR collections
- ▼ **Traffic flow**
 - Traffic policies
 - Policy records
- ▼ **Domains**
 - Registered domains
 - Requests
- ▼ **Resolver**

Hosted zones (1)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

	Hosted zone name	Type	Created by	Record co...	Description	Hosted z...
☐	techtwinklesharma.online	Public	Route 53	14	-	Z0985708...

Click on hosted zones

Click here

<https://www.youtube.com/@techwithtwinkle>

aws twinkl

Route 53 > Hosted zones > techtwinklesharma.online

Public techtwinklesharma.online Info Delete zone Test record Configure query logging

Hosted zone details Edit hosted zone

Records (14) DNSSEC signing Hosted zone tags (0)

Records (14) Info Delete record Import zone file Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias

Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	
☐	techwink...	NS	Simple	-	No	ns-1521.awsdns-62.org. ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk	172800

aws twinkl

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Record 1

Record name Info Failover .techtwinklesharma.online

Keep blank to create a record for the root domain.

Alias

Value Info 13.127.17.56

Enter multiple values on separate lines.

TTL (seconds) Info 300

Recommended values: 60 to 172800 (two days)

Routing policy Info Failover

Health check ID - optional Info

Failover record type

Enter Failover

Enter IP

Select Record A

Select failover

<https://www.youtube.com/@techwithtwinkle>

aws twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Enter multiple values on separate lines.

TTL (seconds) Info
60
Recommended values: 60 to 172800 (two days)

Routing policy Info
Failover

Health check ID Info
9a283bbb-6337-4a17-8c08-0341291a3ab7

Failover record type
Choose **Primary** to route traffic to the specified resource by default or **Secondary** to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.
Primary

Record ID Info
South 1

<https://www.youtube.com/@techwithtwinkle>

Select TTL

Select Location health check

Select primary

Click on ADD

Add another record

aws twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Failover .techtwinklesharma.online A – Routes traffic to an IPv4 address and some AWS resources

Keep blank to create a record for the root domain.

☐ Alias

Value Info
98.130.57.108

Enter multiple values on separate lines.

TTL (seconds) Info
60
Recommended values: 60 to 172800 (two days)

Routing policy Info
Failover

Failover record type
Choose **Primary** to route traffic to the specified resource by default or **Secondary** to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.
Primary

Health check ID Info
2c0911f2-3b19-4d9d-af68-2e98a4469b6b

Record ID Info
South-2

Same as previous only change the IP & Health check

Select record type secondary

Route 53 > Hosted zones > techtwinklesharma.online > Create record

TTL (seconds) Info

60

Recommended values: 60 to 172800 (two days)

1m1h1d

Routing policy Info

Failover

Health check ID - optional Info

2c0911f2-3b19-4d9d-af68-2e98a4469b6b

Failover record type

Choose Primary to route traffic to the specified resource by default or Secondary to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.

Secondary

Record ID Info

South-2

Add another record

<https://www.youtube.com/@techwithtwinkle>

Click on create record



View existing records

The following table lists the existing records in techtwinklesharma.online.

awsSearch[Alt+S]Globalaws-twinkle-v4 (2355-6299-1793)twinkle

Route 53 > Hosted zones > techtwinklesharma.online

Route 53

Dashboard

Hosted zones

Health checks

Profiles New

IP-based routing

CIDR collections

Traffic flow

Traffic policies

Policy records

Domains

Registered domains

Requests

Resolver

VPCs

Inbound endpoints

Records (16) Info

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value

Type

Routing p...

Alias

1

Delete record

Import zone file

Create record

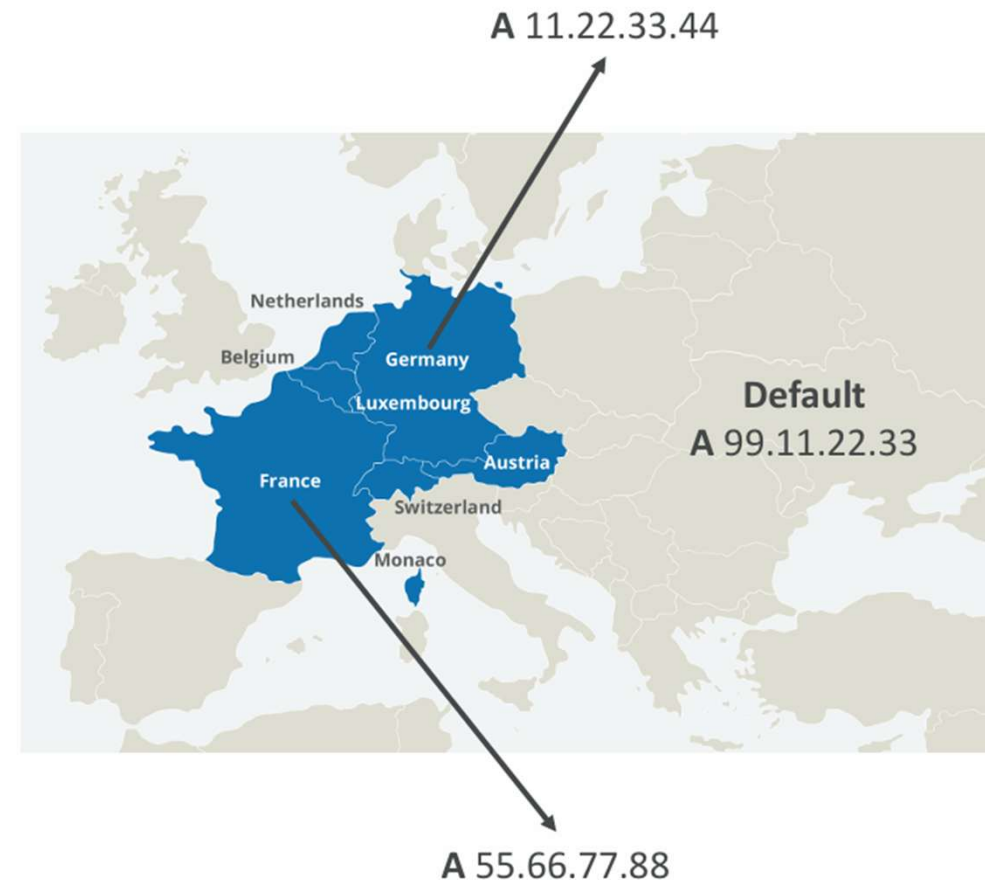
	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...
☐	techtwink...	NS	Simple	-	No	ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk.	172800	-
☐	techtwink...	SOA	Simple	-	No	ns-1521.awsdns-62.org. aws...	900	-
☐	demo.tec...	A	Simple	-	No	56.155.98.247	120	-
☐	failover.te...	A	Failover	Primary	No	13.127.17.56	60	9a283bbb...
☐	failover.te...	A	Failover	Secondary	No	98.130.57.108	60	2c0911f2...
☐	latency.te...	A	Latency	Asia Paci...	No	56.155.98.247	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	13.127.17.56	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	98.130.57.108	300	-
☐	myalias.te...	A	Simple	-	Yes	dualstack.demoroute53alb-1...	-	-
☐	myapp.tec...	CNAME	Simple	-	No	DemoRoute53ALB-1908756...	300	-
☐						56.155.98.247		

Failover
is
created

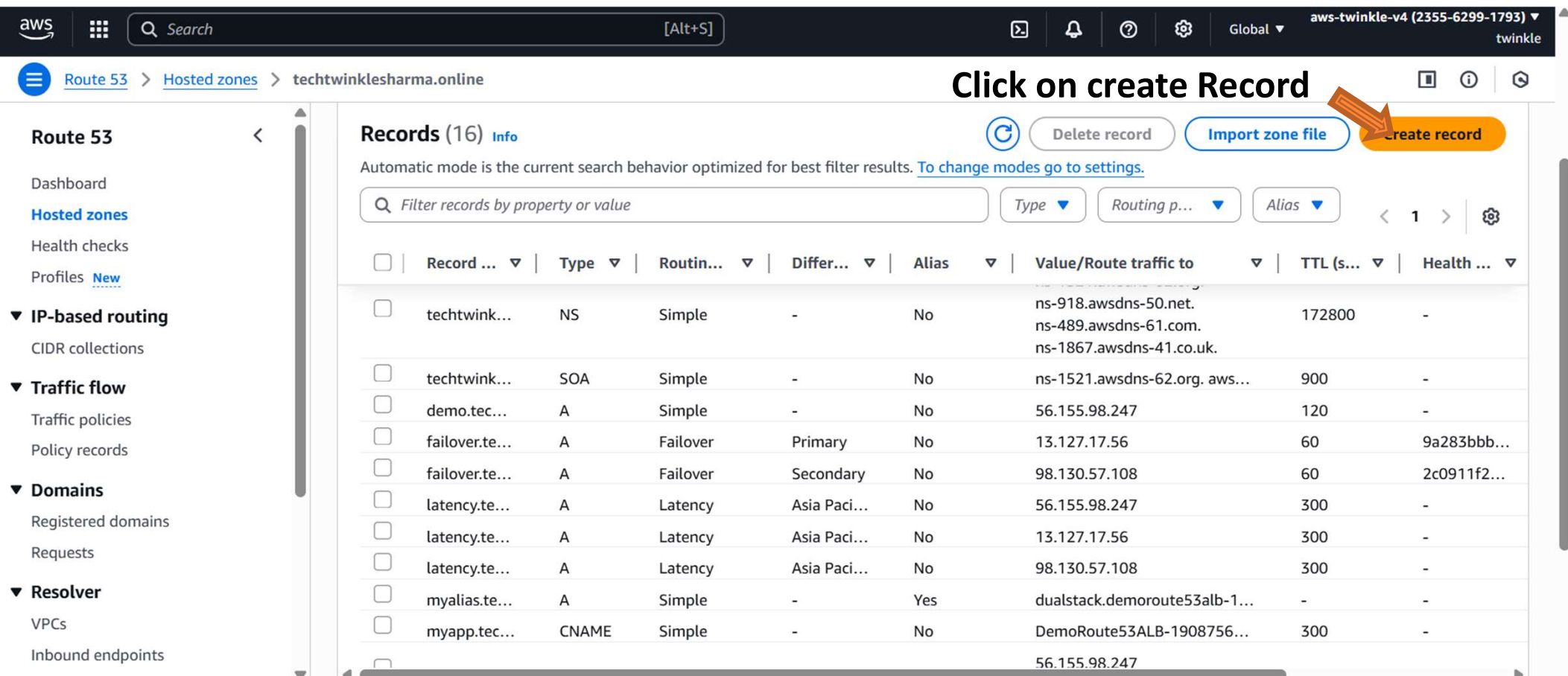


Routing Policies – Geolocation

- ▶ Different from Latency-based!
- ▶ **This routing is based on user location**
- ▶ Specify location by Continent, Country or by US State (if there's overlapping, most precise location selected)
- ▶ Should create a “**Default**” record (in case there's no match on location)
- ▶ Use cases: website localization, restrict content distribution, load balancing, ...
- ▶ Can be associated with Health Checks



Routing Policies – Geolocation Hands On



Click on create Record

Records (16) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

☐	Record ... ▾	Type ▾	Routin... ▾	Differ... ▾	Alias ▾	Value/Route traffic to ▾	TTL (s... ▾	Health ... ▾
☐	techtwink...	NS	Simple	-	No	ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk.	172800	-
☐	techtwink...	SOA	Simple	-	No	ns-1521.awsdns-62.org. aws...	900	-
☐	demo.tec...	A	Simple	-	No	56.155.98.247	120	-
☐	failover.te...	A	Failover	Primary	No	13.127.17.56	60	9a283bbb...
☐	failover.te...	A	Failover	Secondary	No	98.130.57.108	60	2c0911f2...
☐	latency.te...	A	Latency	Asia Paci...	No	56.155.98.247	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	13.127.17.56	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	98.130.57.108	300	-
☐	myalias.te...	A	Simple	-	Yes	dualstack.demoroute53alb-1...	-	-
☐	myapp.tec...	CNAME	Simple	-	No	DemoRoute53ALB-1908756...	300	-
☐						56.155.98.247		

<https://www.youtube.com/@techwithtwinkle>

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Info

Create record

Info

Quick create record

▼ Record 1

Record name

Info

geo

.techtwinklesharma.online

Record type

Info

A – Routes traffic to an IPv4 address and some AWS resources

Delete

Alias

Value

Info

56.155.98.247

TTL (seconds)

Info

300

1m

1h

1d

Routing policy

Info

Geolocation

Enter multiple values on separate lines.

Enter name

Select record type

Paste IP of Instance

Select Geolocation

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Info

TTL (seconds)

Info

300

1m

1h

1d

Routing policy

Info

Geolocation

Location

Choose a location that DNS queries originate from. Queries from that location are routed to the resource that you specify in this record. Choose **Default** to route queries from locations that you don't create records for.

Asia

Health check ID - optional

Info

Choose health check

Refresh

Record ID

Info

Asia

Select Asia

Enter Asia

Click on Add Another record

Add another record

Cancel

Create records

View existing records

The following table lists the existing records in techtwinklesharma.online.

<https://www.youtube.com/@techwithtwinkle>

Record ID Info

Asia

▼ Record 2

Delete

Record name Info

geo

.techtwinklesharma.online

Record type Info

A – Routes traffic to an IPv4 address and some AWS resources

☐ Alias

Value Info

98.130.57.108

Enter multiple values on separate lines.

TTL (seconds) Info

300

Recommended values: 60 to 172800 (two days)

1m

1h

1d

Routing policy Info

Geolocation

Location

Health check ID - optional Info



Search

[Alt+S]



Global

aws-twinkle-v4 (2355-6299-1793) twinkle

TTL (seconds) Info

300

Recommended values: 60 to 172800 (two days)

1m

1h

1d

Routing policy Info

Geolocation

Location

Choose a location that DNS queries originate from. Queries from that location are routed to the resource that you specify in this record. Choose **Default** to route queries from locations that you don't create records for.

United States

Health check ID - optional Info

Choose health check

Select Country

Record ID Info

us

Click on Add record

Add another record

Cancel

Create records

<https://www.youtube.com/@techwithtwinkle>

► View existing records

The following table lists the existing records in techtwinklesharma.online.

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Record ID

Info

US

▼ Record 3

Record name

Info

geo

.techtwinklesharma.online

Record type

Info

A – Routes traffic to an IPv4 address and some AWS resources

Delete

Keep blank to create a record for the root domain.

Alias

Value

Info

13.127.17.56

Enter multiple values on separate lines.

TTL (seconds)

Info

300

1m

1h

1d

Routing policy

Info

Geolocation

Recommended values: 60 to 172800 (two days)

Location

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Enter multiple values on separate lines.

https://www.youtube.com/@techwithtwinkle

TTL (seconds)

Info

300

1m

1h

1d

Routing policy

Info

Geolocation

Recommended values: 60 to 172800 (two days)

Location

Choose a location that DNS queries originate from. Queries from that location are routed to the resource that you specify in this record. Choose **Default** to route queries from locations that you don't create records for.

Default

Health check ID - optional

Info

Choose health check

Record ID

Info

Default south

Enter here

Select default

Add another record

Click here to create records

Create records

► View existing records

aws

Search

[Alt+S]

Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

Hosted zones

techtwinklesharma.online

Route 53

Dashboard

Hosted zones

Health checks

Profiles

New

IP-based routing

CIDR collections

Traffic flow

Traffic policies

Policy records

Domains

Registered domains

Requests

Resolver

VPCs

Inbound endpoints

Records (19)

Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type

Routing p...

Alias

1

	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...
☐	demo.tec...	A	Simple	-	No	56.155.98.247	120	-
☐	failover.te...	A	Failover	Primary	No	13.127.17.56	60	9a283bbb...
☐	failover.te...	A	Failover	Secondary	No	98.130.57.108	60	2c0911f2...
☐	geo.techt...	A	Geolocation	Asia	No	56.155.98.247	300	-
☐	geo.techt...	A	Geolocation	Default	No	13.127.17.56	300	-
☐	geo.techt...	A	Geolocation	United S...	No	98.130.57.108	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	56.155.98.247	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	13.127.17.56	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	98.130.57.108	300	-
☐	myalias.te...	A	Simple	-	Yes	dualstack.demoroute53alb-1...	-	-
☐	myapp.tec...	CNAME	Simple	-	No	DemoRoute53ALB-1908756...	300	-
☐	simple.tec...	A	Simple	-	No	56.155.98.247	20	-
						98.130.57.108		

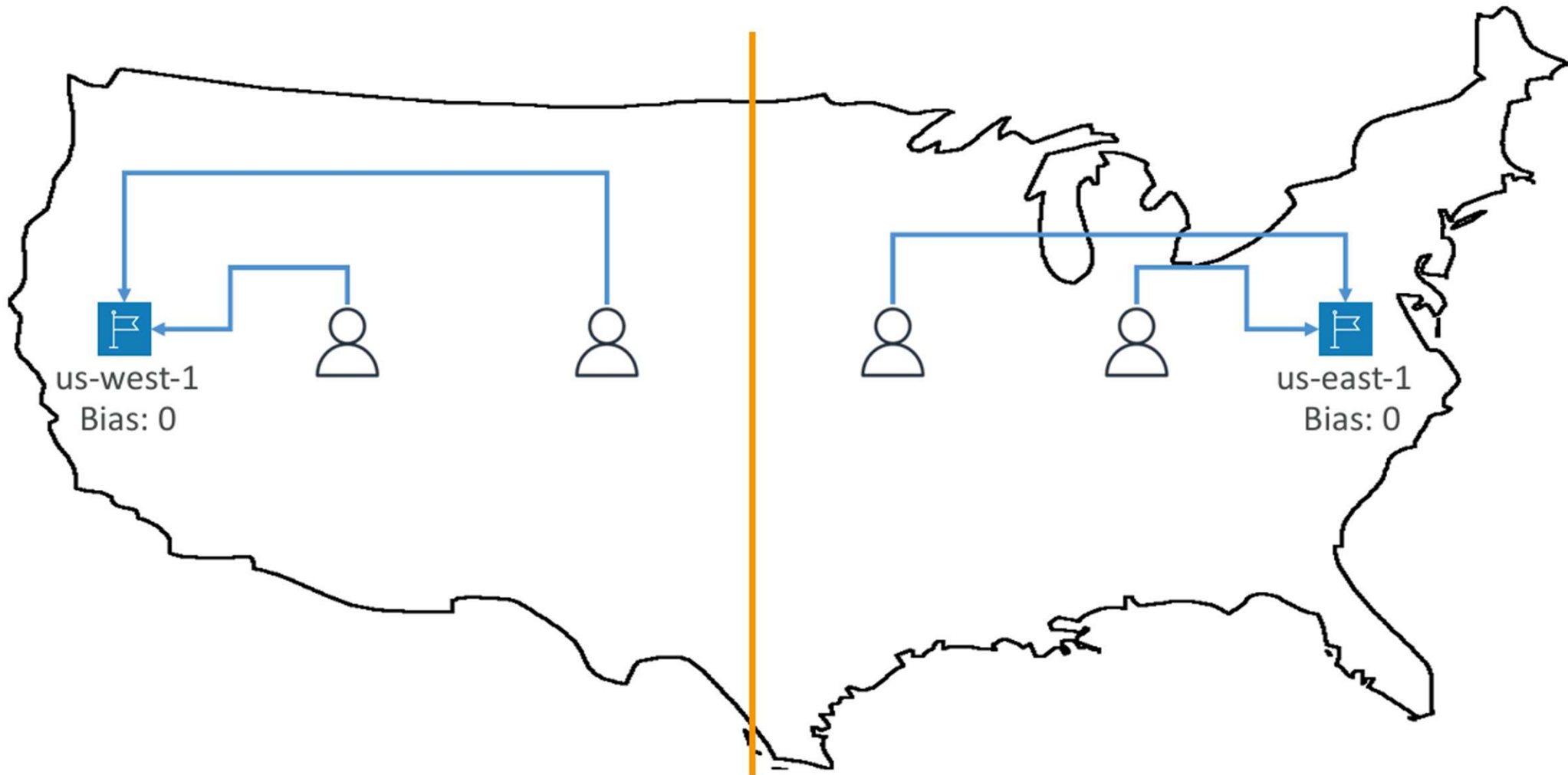
Geolocation
created

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Routing Policies – Geoproximity

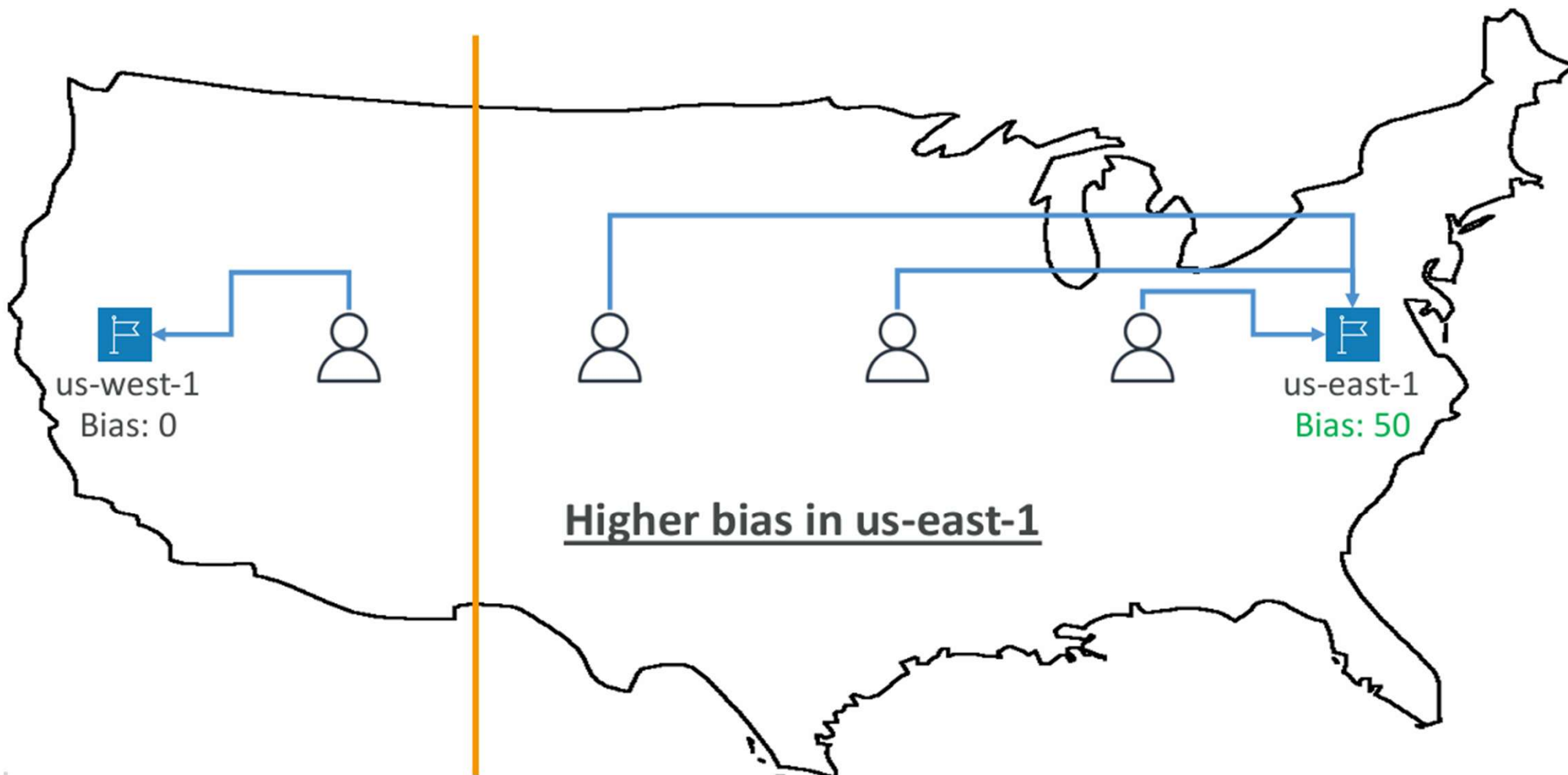
- ▶ Route traffic to your resources based on the geographic location of users and resources
- ▶ Ability **to shift more traffic to resources based** on the defined bias
- ▶ To change the size of the geographic region, specify bias values:
 - ▶ To expand (1 to 99) – more traffic to the resource
 - ▶ To shrink (-1 to -99) – less traffic to the resource
- ▶ Resources can be:
 - AWS resources (specify AWS region)
 - Non-AWS resources (specify Latitude and Longitude)
- ▶ You must use Route 53 **Traffic Flow** to use this feature

Routing Policies – Geoproximity



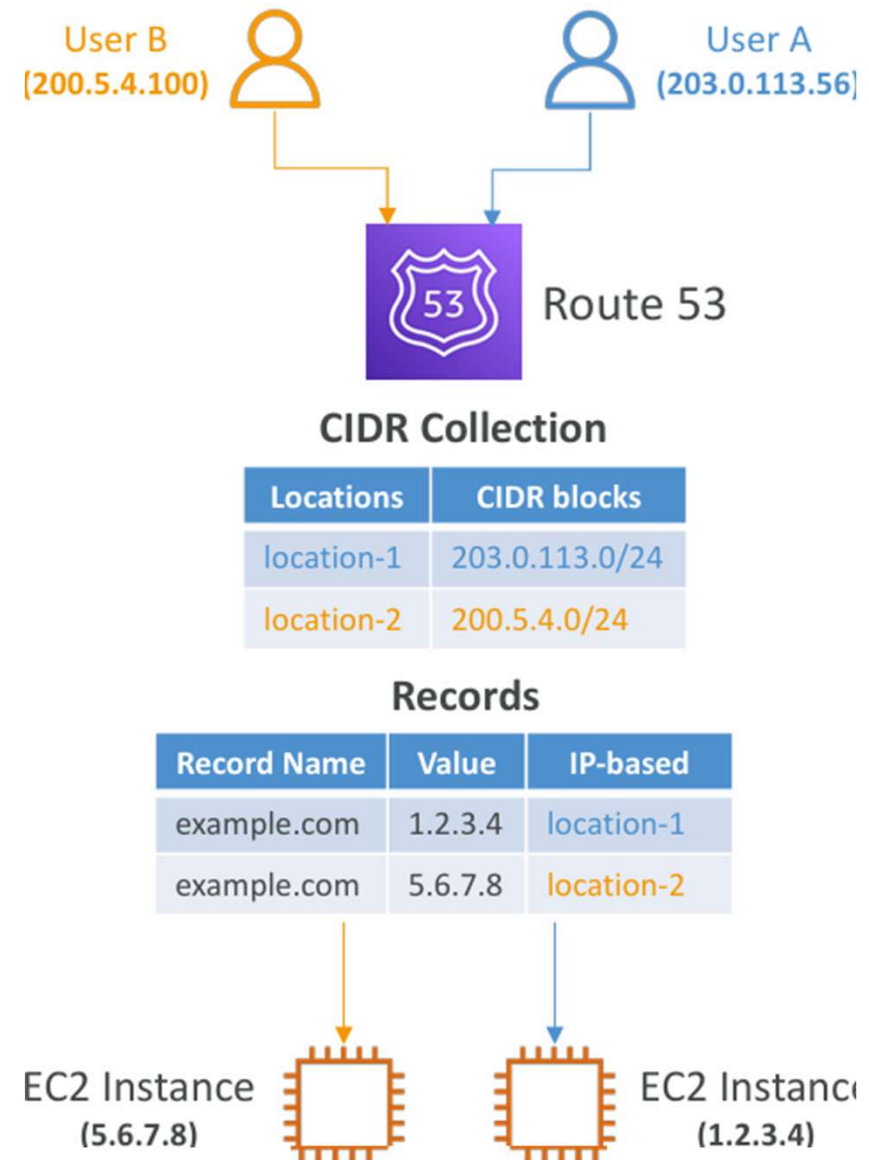
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Routing Policies – Geoproximity



Routing Policies – IP-based Routing

- ▶ Routing is based on clients' IP addresses
- ▶ You provide a **list of CIDRs** for **your clients** and the corresponding endpoints/locations (user-IP-to-endpoint mappings)
- ▶ Use cases: Optimize performance, reduce network costs...
- ▶ Example: route end users from a particular ISP to a specific endpoint



Routing Policies – Multi-Value

- ▶ Use when routing traffic to multiple resources
- ▶ Route 53 return multiple values/resources
- ▶ Can be associated with Health Checks (return only values for healthy resources)
- ▶ Up to 8 healthy records are returned for each Multi-Value query
- ▶ **Multi-Value is not a substitute for having an ELB**

Name	Type	Value	TTL	Set ID	Health Check
www.example.com	A Record	192.0.2.2	60	Web1	A
www.example.com	A Record	198.51.100.2	60	Web2	B
www.example.com	A Record	203.0.113.2	60	Web3	C

Routing Policies – Multi-Value Hands On

The screenshot shows the AWS Route 53 console interface. The left sidebar contains navigation links for Route 53, Hosted zones, Health checks, Profiles, IP-based routing, Traffic flow, Domains, and Resolver. The main content area displays a list of 19 DNS records for the hosted zone techtwinklesharma.online. An orange arrow points to the 'Create record' button in the top right corner of the records list.

Records (19) [Info](#) **Click on create record** [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type ▼ Routing p... ▼ Alias ▼ < 1 > ⚙

☐	Record ... ▼	Type ▼	Routin... ▼	Differ... ▼	Alias ▼	Value/Route traffic to ▼	TTL (s... ▼	Health ... ▼
☐	techtwink...	NS	Simple	-	No	ns-918.awsdns-50.net. ns-489.awsdns-61.com. ns-1867.awsdns-41.co.uk.	172800	-
☐	techtwink...	SOA	Simple	-	No	ns-1521.awsdns-62.org. aws...	900	-
☐	demo.tec...	A	Simple	-	No	56.155.98.247	120	-
☐	failover.te...	A	Failover	Primary	No	13.127.17.56	60	9a283bbb...
☐	failover.te...	A	Failover	Secondary	No	98.130.57.108	60	2c0911f2...
☐	geo.techt...	A	Geolocation	Asia	No	56.155.98.247	300	-
☐	geo.techt...	A	Geolocation	Default	No	13.127.17.56	300	-
☐	geo.techt...	A	Geolocation	United S...	No	98.130.57.108	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	56.155.98.247	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	13.127.17.56	300	-
☐	latencv.te...	A	Latencv	Asia Paci...	No	98.130.57.108	300	-

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Global

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twinkle

Route 53

Hosted zones

techtwinklesharma.online

Create record

Info

Quick create record

Record 1

Record name

multi

.techtwinklesharma.online

Record type

A – Routes traffic to an IPv4 address and some AWS resources

Switch to wizard

Delete

Enter record name

Enter IP

Select A

Value

98.130.57.108

Enter multiple values on separate lines.

TTL (seconds)

300

Recommended values: 60 to 172800 (two days)

Routing policy

Keep blank to create a record for the root domain.

Alias

You can't use an alias record with multivalue routing policy.

Value

98.130.57.108

Enter multiple values on separate lines.

TTL (seconds)

300

Recommended values: 60 to 172800 (two days)

Routing policy

Multivalue answer

Health check ID - optional

Choose health check

Record ID

US West load balancer

Add another record

Cancel

Create records

aws

Search

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Global

aws-twinkle-v4 (2355-6299-1793)

twinkle

Route 53

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techtwinklesharma.online

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300

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Routing policy

Multivalue answer

Health check ID - optional

Choose health check

Record ID

US West load balancer

Add another record

Cancel

Create records

Click on this

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Select Multivalue answer

Keep blank to create a record for the root domain.

☐ Alias
You can't use an alias record with multivalue routing policy.

Value [Info](#)

98.130.57.108

-

ap-northeast-3

ID: 222d366a-d344-40bc-b24d-323a8a230c65

ap-south-1

ID: 9a283bbb-6337-4a17-8c08-0341291a3ab7

ap-south-2

ID: 2c0911f2-3b19-4d9d-af68-2e98a4469b6b

calculated

ID: 955ff86a-ccda-4b95-b9d4-8b5278cf97f3

[Choose health check](#)

ap-south-2

Routing policy [Info](#)

Multivalue answer

Record ID [Info](#)

US West load balancer

Select ap-south-2

[Add another record](#)

[Cancel](#)

[Create records](#)

Keep blank to create a record for the root domain.

☐ Alias
You can't use an alias record with multivalue routing policy.

Value [Info](#)

98.130.57.108

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

60

Recommended values: 60 to 172800 (two days)

Health check ID - optional [Info](#)

[2c0911f2-3b19-4d9d-af68-2e98a4469b6b](#)

Routing policy [Info](#)

Multivalue answer

Record ID [Info](#)

south-2

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Enter 60

Click on Add another record

[Add another record](#)

[Cancel](#)

[Create records](#)

aws twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

Health check ID - optional [Info](#)

Record ID [Info](#)

2c0911f2-3b19-4d9d-af68-2e98a4469b6b south-2

▼ Record 2 [Delete](#)

Record name [Info](#) **Enter Name**

multi .techtwinklesharma.online

Record type [Info](#)

A - Routes traffic to an IPv4 address and some AWS resources

Keep blank to create a record for the root domain.

☐ Alias
You can't use an alias record with multivalue routing policy.

Value [Info](#) **Enter IP of another Instance**

56.155.98.247

Enter multiple values on separate lines.

TTL (seconds) [Info](#) **Enter 60**

60 1m 1h 1d

Routing policy [Info](#) **Select multivalue answer**

Multivalue answer

Recommended values: 60 to 172800 (two days)

aws twinkle-v4 (2355-6299-1793) twinkle

Route 53 > Hosted zones > techtwinklesharma.online > Create record

56.155.98.247

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Enter multiple values on separate lines.

TTL (seconds) [Info](#)

60 1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Multivalue answer

Health check ID - optional [Info](#) **Select health check**

222d366a-d344-40bc-b24d-323a8a230c65

Record ID [Info](#)

North east

[Add another record](#) **Click on Add Another record**

[Cancel](#) [Create records](#)

► View existing records

The following table lists the existing records in techtwinklesharma.online.

Health check ID - optional Info

222d366a-d344-40bc-b24d-323a8a230c65

Record ID Info

North east

Record 3

Enter name

Record name Info

multi

.techtwinklesharma.online

Record type Info

A - Routes traffic to an IPv4 address and some AWS resources

Delete

Keep blank to create a record for the root domain.

Alias

You can't use an alias record with multivalue routing policy.

Value Info

13.127.17.56

Enter IP of left instance

Enter multiple values on separate lines.

TTL 60

TTL (seconds) Info

60

1m

1h

1d

Routing policy Info

Multivalue answer

Select multivalue answer

Recommended values: 60 to 172800 (two days)

Keep blank to create a record for the root domain.

Alias

You can't use an alias record with multivalue routing policy.

Value Info

13.127.17.56

Enter multiple values on separate lines.

TTL (seconds) Info

60

1m

1h

1d

Routing policy Info

Multivalue answer

Health check ID - optional Info

9a283bbb-6337-4a17-8c08-0341291a3ab7

Record ID Info

South 1

Select location health check of that instance

Click on create record

Add another record

Create records

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Route 53 > Hosted zones > techtwinklesharma.online

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles [New](#)

▼ **IP-based routing**

- CIDR collections

▼ **Traffic flow**

- Traffic policies
- Policy records

▼ **Domains**

- Registered domains
- Requests

▼ **Resolver**

- VPCs
- Inbound endpoints

Records (22) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type Routing p... Alias

1

	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...
☐	geo.tech...	A	Geolocation	Default	No	13.127.17.56	300	-
☐	geo.tech...	A	Geolocation	United S...	No	98.130.57.108	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	56.155.98.247	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	13.127.17.56	300	-
☐	latency.te...	A	Latency	Asia Paci...	No	98.130.57.108	300	-
☐	multi.tech...	A	Multivalu...	-	No	56.155.98.247	60	222d366a...
☐	multi.tech...	A	Multivalu...	-	No	13.127.17.56	60	9a283bbb...
☐	multi.tech...	A	Multivalu...	-	No	98.130.57.108	60	2c0911f2...
☐	myalias.te...	A	Simple	-	Yes	dualstack.demoroute53alb-1...	-	-
☐	myapp.tec...	CNAME	Simple	-	No	DemoRoute53ALB-1908756...	300	-
☐	simple.tec...	A	Simple	-	No	56.155.98.247 98.130.57.108	20	-
☐	test.tech...	A	Simple	-	No	11.22.33.44	300	-

Multi created

After all the practice delete hosted zones, instances & health checks

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Domain Registrar vs. DNS Service

- ▶ You buy or register your domain name with a Domain Registrar typically by paying annual charges (e.g., GoDaddy, Amazon Registrar Inc., ...)
- ▶ The Domain Registrar usually provides you with a DNS service to manage your DNS records
- ▶ But you can use another DNS service to manage your DNS records
- ▶ Example: purchase the domain from GoDaddy and use Route 53 to manage your DNS records



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GoDaddy as Registrar & Route 53 as DNS Service



Records

We can't display your DNS information because your nameservers aren't managed by us.

Nameservers

Using custom nameservers [Change](#)

Nameserver

ns-1083.awsdns-07.org

ns-932.awsdns-52.net

ns-1911.awsdns-46.co.uk

ns-481.awsdns-60.com



Amazon
Route 53

Public Hosted Zone
stephanetheteacher.com

▼ Hosted zone details [Edit hosted zone](#)

Hosted zone ID Z30IUJCCWPKZUV	Type Public hosted zone	Name servers ns-252.awsdns-31.com ns-1468.awsdns-55.org ns-633.awsdns-15.net ns-1800.awsdns-33.co.uk
Description HostedZone created by Route53 Registrar	Record count 22	
Query log		

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3rd Party Registrar with Amazon Route 53

- ▶ **If you buy your domain on a 3rd party registrar, you can still use Route 53 as the DNS Service provider**
- ▶ Create a Hosted Zone in Route 53 2
- ▶ Update NS Records on 3rd party website to use Route 53 **Name Servers**
- ▶ **Domain Registrar != DNS Service**
- ▶ But every Domain Registrar usually comes with some DNS features

Route 53 Hybrid DNS

- ▶ **Route 53 Hybrid DNS** is when you combine **Amazon Route 53** with your **on-premises DNS** so that DNS queries can resolve **both AWS-hosted and on-premises domain names**. It's used in **hybrid cloud architectures** where part of your infrastructure is in AWS and part remains on-premises (or in another cloud).
- ▶ **Why you need it**
- ▶ If you have:
- ▶ **AWS resources** (like EC2 instances, RDS, S3 static websites) that should be resolved privately.
- ▶ **On-premises resources** (like internal apps, file servers) that must also be resolved by AWS-based workloads.
- ▶ A **single, unified DNS resolution path** so workloads can find each other regardless of where they are hosted.

Key Components

Component	Purpose
Private Hosted Zone	AWS-only domain namespace
Inbound Endpoint	Lets on-prem query AWS private zones
Outbound Endpoint	Lets AWS query on-prem DNS zones
Resolver Rules	Define which domains go where
Forwarding	On-prem DNS config to send AWS domain queries to AWS inbound endpoint

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Resolver Endpoints

Resolver Endpoints in **Amazon Route 53** are the building blocks that let your AWS and on-premises networks talk to each other's DNS systems in a hybrid cloud setup.

They are part of **Route 53 Resolver** (the DNS service inside AWS VPCs) and come in **two types**:

Types

1. Inbound Resolver Endpoint

Purpose: Lets **on-premises DNS servers** resolve **AWS private DNS names**.

Direction: From on-prem → AWS.

Example use case:

On-prem app wants to connect to db.aws.internal (hosted in AWS Private Hosted Zone).

How it works:

On-prem DNS receives a query for db.aws.internal.

It forwards the query to the **inbound endpoint** in AWS (IP in VPC).

Route 53 Resolver answers with the internal AWS resource's IP.

2. Outbound Resolver Endpoint

Purpose: Lets **AWS workloads** resolve **on-premises DNS names**.

Direction: From AWS → on-prem.

Example use case:

An EC2 instance in AWS needs to connect to fileserver.corp.local (on-prem domain).

How it works:

EC2 queries Route 53 Resolver.

Resolver rule sends the query to the outbound endpoint.

Outbound endpoint forwards it to the on-prem DNS server.

On-prem DNS replies with the IP.

Key Details

Feature	Inbound Endpoint	Outbound Endpoint
Traffic Direction	On-prem → AWS	AWS → On-prem
Use Case	Resolve AWS private zones from on-prem	Resolve on-prem zones from AWS
Requires Resolver Rules?	No	Yes
IP Addresses	Elastic Network Interfaces in your VPC	Elastic Network Interfaces in your VPC
Connectivity	VPN or Direct Connect (Private VIF)	VPN or Direct Connect (Private VIF)

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Resolver Rules

Resolver Rules in Amazon Route 53 control **where DNS queries go** when using Route 53 Resolver — especially for hybrid DNS setups.

They're basically **routing rules for DNS queries**.

Why Resolver Rules Exist

AWS's default VPC DNS can resolve:

- Public DNS names** (internet)

- Private Hosted Zones** (in AWS)

But if you need to resolve **on-premises domains** (e.g., corp.local) or send some queries to another DNS service, you need to tell Route 53 **where to forward those queries**. That's where resolver rules come in.

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Types of Resolver Rules

Forwarding Rules

Forward queries for specific domains to a target DNS server (on-prem or elsewhere).

Example: Forward corp.local queries to **on-prem DNS** at 10.0.0.2.

System Rules (Default)

Automatically resolve public domains via AWS's default DNS.
No manual setup required.

Conditional Forwarding Rules

Forward only certain domains to certain endpoints.

Example: Forward aws.internal to the **inbound endpoint** from on-prem.

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Key Rule Components

Component	Description
Domain Name	The suffix the rule applies to (e.g., <code>corp.local</code>)
Rule Type	Forward or System
Target IP(s)	IP addresses of the DNS servers to forward queries to
Associated VPCs	The VPCs where this rule is applied
Direction	Always applies to outbound queries from AWS (inbound queries don't need rules)

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